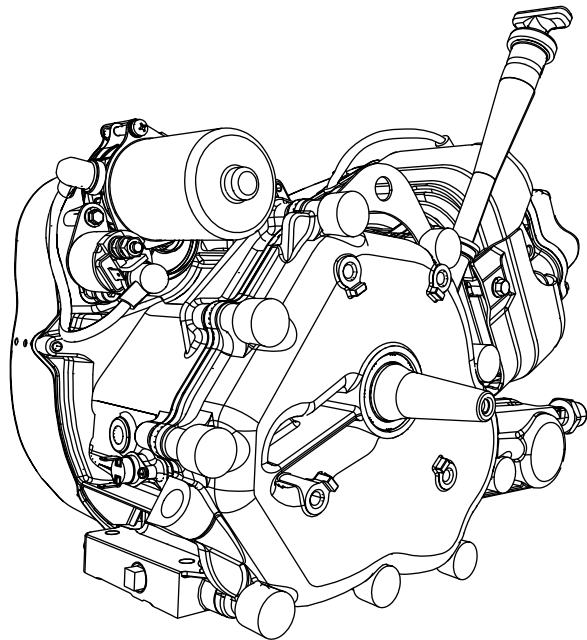


Generac 426/460 Single Cylinder Engine

Service Manual



SAVE THIS MANUAL FOR FUTURE REFERENCE

 WARNING

CANCER AND REPRODUCTIVE HARM

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(000393a)

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Foreword

Introduction

This manual has been written and published by Generac Power Systems, Inc. ("Generac") to aid dealers' mechanics, company service personnel, and general consumers when servicing products described herein.

It is assumed these personnel are familiar with servicing procedures for these products, or similar products, manufactured and marketed by Generac. It is also assumed they have been trained in the recommended servicing procedures for these products, which includes the use of mechanics' hand tools and any special tools required.

Correct service and repair is important to the safe, economical, and reliable operation of the products described herein. Troubleshooting, testing, service, and repair procedures recommended by Generac and described in this manual are effective methods of performing such operations. Some of these operations or procedures may require use of specialized equipment. Such equipment should be used when and as recommended.

Generac could not possibly know of and advise the service trade of all conceivable procedures or methods by which a service might be performed, nor of any possible hazards and/or results of each procedure or method. Generac has not undertaken any such wide evaluation. Therefore, anyone who uses a procedure or method not recommended by the manufacturer must first satisfy themselves that neither their safety, nor the product's safety, will be endangered by the service or operating procedure selected.

Some components or assemblies of the product described in this manual may not be considered repairable. Disassembly, repair, and reassembly of such components may not be included in this manual.

Engines described herein may be used to power a wide variety of products. Service and repair instructions relating to any such products are not covered in this manual. For information pertaining to use of these engines with other products, see owner's or service manuals pertaining to said products.

The information in this manual is accurate based on products produced at the time of publication. The manufacturer reserves the right to make technical updates, corrections, and product revisions at any time without notice.

A Word About Special Tools

Many of the procedures depicted in this manual require the use of special tools. Some of the tools required are available as Generac parts and are listed as such in this manual.

Intended Use

This engine was designed and manufactured for specific applications. Do not attempt to modify the equipment or use it for any application for which it was not designed.

Generac Power Systems, Inc. does not approve or authorize use of these engines on All Terrain Vehicles (ATVs), go-carts, motorbikes, aircraft products, personal watercraft, or vehicles intended for use in competitive events. Use of this product in any other than its intended application will void the warranty! Use of these engines in such applications could result in property damage, serious injury (including paralysis), or even death.

Service Information

Service on this engine within and after the warranty period can be performed by any Independent Authorized Service Dealer (IASD). Service technicians are factory trained and capable of handling all service needs.

Always supply the complete model number and serial number of your unit as given on its data plate decal when contacting an IASD about parts and service.

Warranty statement for this engine is included with the documents provided with the unit.

Scope

Servicing information provided in this manual applies only to the Generac engine itself. Topics such as air cleaner maintenance, external governor adjustment, and carburetor overhaul are not included. For servicing information on those components, see diagnostic manual for the associated Generac product.

Servicing procedures apply to engines used in both portable and home standby (HSB) equipment. Two sets of instructions and illustrations are provided where significant design differences exist.

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Section 1: Safety Rules & General Information

Notice to Users of this Manual

Study these safety rules carefully before operating or servicing this equipment. Become familiar with the owner's manual and with the engine. The engine can operate safely, efficiently, and reliably only if it is correctly operated and maintained. Many accidents are caused by failing to follow simple and fundamental rules or precautions.

The information in this manual is accurate based on products produced at the time of publication. The manufacturer reserves the right to make technical updates, corrections, and product revisions at any time without notice.

Read This Manual Thoroughly



Consult Manual. Read and understand manual completely before using product. Failure to completely understand manual and product could result in death or serious injury. (000100a)

If any section of this manual is not understood, contact the nearest Independent Authorized Service Dealer (IASD) or Generac Customer Service at 1-888-436-3722 (1-888-GENERAC), or visit www.generac.com for starting, operating, and servicing procedures. The owner is responsible for correct maintenance and safe use of the unit.

This manual must be used in conjunction with all other supporting product documentation supplied with the product.

SAVE THESE INSTRUCTIONS for future reference. This manual contains important instructions that must be followed during placement, operation, and maintenance of the unit and its components. Always supply this manual to any individual that will use this unit, and instruct them on how to correctly start, operate, and stop the unit in case of emergency.

Safety Rules

The manufacturer cannot anticipate every possible circumstance that might involve a hazard. The alerts in this manual, and on tags and decals affixed to the unit, are not all inclusive. If using a procedure, work method, or operating technique that the manufacturer does not specifically recommend, verify that it is safe for others and does not render the equipment unsafe.

Throughout this publication, and on tags and decals affixed to the unit, DANGER, WARNING, CAUTION, and NOTE blocks are used to alert personnel to special

instructions about a particular operation that may be hazardous if performed incorrectly or carelessly. Observe them carefully. Alert definitions are as follows:



Indicates a hazardous situation which, if not avoided, will result in death or serious injury.

(000001)



Indicates a hazardous situation which, if not avoided, could result in death or serious injury.

(000002)



Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.

(000003)

NOTE: Notes contain additional information important to a procedure and will be found within the regular text of this manual.

These safety alerts cannot eliminate the hazards that they indicate. Common sense and strict compliance with the special instructions while performing the action or service are essential to preventing accidents.

General Safety Information

WARNING

Personal injury. Never remove, alter, or tamper with safety devices. Do not operate unit without safety equipment in place. Doing so could result in death, serious injury, property damage, or equipment damage.

(000403)

WARNING

Equipment damage. Do not attempt to start or operate a unit in need of repair or scheduled maintenance. Doing so could result in serious injury, death, or equipment failure or damage.

(000291)

WARNING

Risk of injury. Do not operate or service this machine if not fully alert. Fatigue can impair the ability to service this equipment and could result in death or serious injury.

(000215)

IMPORTANT NOTE: Incorrect treatment of this engine can damage it and shorten its life.

- If you have questions about intended use, contact an IASD.

Replacement Parts

This equipment is designed to be used with Generac authorized parts ONLY. If equipment is used with parts that DO NOT comply with minimum specifications, the user assumes all risks and liabilities. During any maintenance procedure, replacement fasteners must have the same measurements and strength as the removed fasteners. Metric bolts and nuts have numbers indicating their strength. Customary bolts use radial lines to indicate strength, while most customary nuts do not have strength markings. Mismatched or incorrect fasteners can cause damage, malfunction, and possible injury.

Exhaust Hazards



DANGER

Asphyxiation. Running engines produce carbon monoxide, a colorless, odorless, poisonous gas. Carbon monoxide, if not avoided, will result in death or serious injury.

(000103)



DANGER

Asphyxiation. Carbon monoxide can kill in minutes. Operate this unit outdoors only. Failure to do so will cause death or serious injury.

(000525)



WARNING

Asphyxiation. Always use a battery operated carbon monoxide alarm indoors and installed according to the manufacturer's instructions. Failure to do so could result in death or serious injury.

(000178a)



WARNING

Asphyxiation. Do not inhale vapor or fumes. Seek medical attention immediately. Failure to do so could result in death or serious injury.

(000342)

CAUTION

Personal injury. Wear appropriate personal protective equipment at all times while operating and servicing unit. Failure to do so could result in personal injury.

(000419)

- Breathing carbon monoxide can cause headache, fatigue, dizziness, vomiting, confusion, seizures, nausea, fainting, or death. IMMEDIATELY move to fresh air if experiencing any symptoms listed above. Seek medical attention immediately.
- Keep exhaust gas from entering a confined area through windows, doors, ventilation intakes, or other openings. Warn any occupants inside about the symptoms of carbon monoxide so they know to move to fresh air if they begin to feel ill.
- Use a respirator or mask whenever there is a possibility of inhaling harmful gas or vapors.
- Read all instructions with respirator or mask and verify the respirator or mask will provide the necessary protection against inhaling harmful gas or vapors. IMMEDIATELY move to fresh air if experiencing any symptoms listed above.

Electrical Safety



⚠ DANGER

Fire hazard. Do not operate unit if connected electrical devices overheat, electrical output is lost, if unit sparks, or if flame or smoke are observed while unit is operating. Doing so will result in death, serious injury, or property or equipment damage.

(000417)



⚠ DANGER

Electrocution. Water contact with a power source, if not avoided, will result in death or serious injury.

(000104)

⚠ WARNING

Personal injury. Never operate in rain or wet conditions. Doing so could result in death or serious injury.

(000618)

- Never service this engine or handle any electrical device while standing in water, while barefoot, or while hands or feet are wet. DANGEROUS ELECTRIC SHOCK will result.
- Never operate the engine in any enclosed compartment or if the engine speed changes.

Fire and Fuel Hazards



⚠ DANGER

Explosion and Fire. Fuel and vapors are extremely flammable and explosive. Add fuel in a well ventilated area. Keep fire and spark away. Failure to do so will result in death or serious injury.

(000105)



⚠ DANGER

Explosion and fire. Fuel and vapors are extremely flammable and explosive. No leakage of fuel is permitted. Keep fire and spark away. Failure to do so will result in death or serious injury.

(000192)



⚠ DANGER

Explosion and Fire. Fuel and vapors are extremely flammable and explosive. Store fuel in a well ventilated area. Keep fire and spark away. Failure to do so will result in death or serious injury.

(000143)



⚠ DANGER

Explosion and Fire. Do not fill fuel tank past full line. Allow for fuel expansion. Overfilling may cause fuel to spill onto engine causing fire or explosion, which will result in death or serious injury.

(000214)



⚠ DANGER

Risk of fire. Allow fuel spills to completely dry before starting engine. Failure to do so will result in death or serious injury.

(000174)



⚠ DANGER

Explosion and Fire. Drain fuel in a well ventilated area, into an approved container only after engine has cooled. Failure to do so will result in death or serious injury.

(000180)



⚠ WARNING

Explosion and Fire. Do not smoke while refueling unit. Failure to do so could result in death, serious injury, or property or equipment damage.

(000284a)



⚠ WARNING

Fire risk. Fuel and vapors are extremely flammable. Do not operate indoors. Doing so could result in death, serious injury, or property or equipment damage.

(000281)

⚠ CAUTION

Engine Damage. Do not operate with incorrect fuel. Doing so will result in reduced service life of the engine.

(000569)

- Avoid spilling gasoline on a hot engine. Comply with all of the laws regulating storage and handling of gasoline.
- It is a violation of California Public Resource Code, Section 4442, to use or operate the engine on any forest-covered, brush-covered, or grass-covered land unless the exhaust system is equipped with a spark arrester, as defined in Section 4442, maintained in effective working order. Other states or federal jurisdictions may have similar laws. Contact the original equipment manufacturer, retailer, or an IASD to obtain a spark arrester designed for the exhaust system installed on this engine.

Hearing Loss Hazard



⚠️ WARNING

Hearing Loss. Hearing protection is recommended when using this machine. Failure to wear hearing protection could result in permanent hearing loss. (000107)

Moving Parts Hazard



⚠️ WARNING

Moving Parts. Keep clothing, hair, and appendages away from moving parts. Failure to do so could result in death or serious injury. (000111)



⚠️ WARNING

Moving Parts. Do not wear jewelry when starting or operating this product. Wearing jewelry while starting or operating this product could result in death or serious injury. (000115)

Before Starting Equipment

⚠️ WARNING

Equipment damage. Do not attempt to start or operate a unit in need of repair or scheduled maintenance. Doing so could result in serious injury, death, or equipment failure or damage. (000291)

⚠️ WARNING

Personal injury. Never rotate engine with spark plug removed. Rotating engine with spark plug removed could result in death or serious injury. (000564)

⚠️ CAUTION

Engine damage. Verify proper type and quantity of engine oil prior to starting engine. Failure to do so could result in engine damage. (000135)

- Verify spark plug, muffler, fuel cap, and air cleaner are in place.

Kickback Hazard



⚠️ WARNING

Recoil Hazard. Recoil could retract unexpectedly. Kickback could result in death or serious injury. (000183)

- When starting engine, pull cord slowly until resistance is felt and then pull rapidly to avoid kickback.

When Operating Equipment

⚠️ WARNING

Equipment damage. Operating engine with obstructed air cooling slots could result in death, serious injury, and unit damage. (000255a)



⚠️ WARNING

Fire hazard. Do not obstruct cooling and ventilating airflow around the engine. Inadequate ventilation could result in fire hazard, possible equipment damage, death or serious injury. (000256)

⚠️ WARNING

Personal injury and equipment damage. Do not tamper with engine governed speed. Operating engine too fast or slow increases the risk of injury or permanent engine damage. (000254)

⚠️ CAUTION

Equipment damage. Never stop the engine by moving throttle control lever to CHOKE. Doing so could result in engine damage. (000457)

- DO NOT tip engine or equipment at angle which causes fuel to spill.
- Reflective exhaust heat may damage the fuel tank, causing fire. Keep at least 5 ft (1.3 m) of clearance on all sides of the engine for adequate cooling, maintenance, and servicing.

Hot Surfaces Hazard



⚠ WARNING

Hot Surfaces. When operating machine, do not touch hot surfaces. Keep machine away from combustibles during use. Hot surfaces could result in severe burns or fire.

(000108)

- AVOID hot exhaust gases.
- Allow equipment to cool before touching.

When Transporting or Repairing Equipment

⚠ WARNING

Accidental start-up. Disconnect spark plug wires when working on unit. Failure to do so could result in death or serious injury.

(000141)

- Transport/repair with fuel tank EMPTY.

Spark Hazards

⚠ WARNING

Accidental start-up. Disconnect spark plug wires when working on unit. Failure to do so could result in death or serious injury.

(000141)

When Testing For Engine Spark



⚠ WARNING

Explosion. Turn fuel supply OFF before checking for spark. Failure to do so could result in death or severe injury.

(000333)

⚠ WARNING

Electric shock. Never test ignition spark with the spark plug or spark plug wire(s) removed. Always use an approved spark tester. Failure to do so could result in death, serious injury, or equipment damage.

(000406)

- Use approved spark plug tester.
- Place the controller in MANUAL mode.

Battery Hazards



⚠ WARNING

Explosion. Do not dispose of batteries in a fire. Batteries are explosive. Electrolyte solution can cause burns and blindness. If electrolyte contacts skin or eyes, flush with water and seek immediate medical attention.

(000162)



⚠ WARNING

Explosion. Batteries emit explosive gases while charging. Keep fire and spark away. Wear protective gear when working with batteries. Failure to do so could result in death or serious injury.

(000137a)



⚠ WARNING

Electrical shock. Disconnect battery ground terminal before working on battery or battery wires. Failure to do so could result in death or serious injury.

(000164)



⚠ WARNING

Risk of burns. Batteries contain sulfuric acid and can cause severe chemical burns. Wear protective gear when working with batteries. Failure to do so could result in death or serious injury.

(000138a)

⚠ WARNING

Environmental Hazard. Always recycle batteries at an official recycling center in accordance with all local laws and regulations. Failure to do so could result in environmental damage, death, or serious injury.

(000228)

Always recycle batteries in accordance with local laws and regulations. Contact your local solid waste collection site or recycling facility to obtain information on local recycling processes. For more information on battery recycling, visit the Battery Council International website at: <http://batteryCouncil.org>

General Maintenance

- Inspect the engine periodically. Repair or replace all damaged or faulty parts immediately.
- Inspect fuel system frequently for leaks or damage. Repair or replace any damaged or leaking component immediately. Never attempt to change, alter, or modify the engine fuel system in any way which might affect safety or compliance with applicable codes and standards.

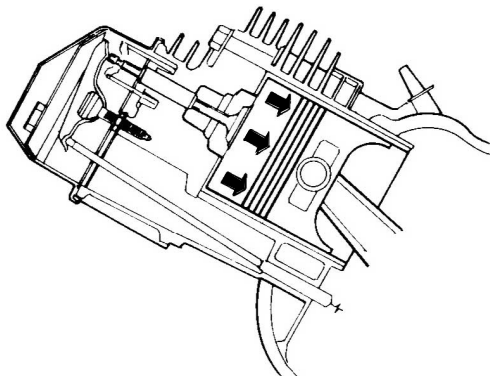
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Section 2: Four-Cycle Engine Theory

Introduction

Four events must occur in the correct sequence and at the correct time for the engine to run correctly. These events are (A) intake, (B) compression, (C) ignition and power, and (D) exhaust.

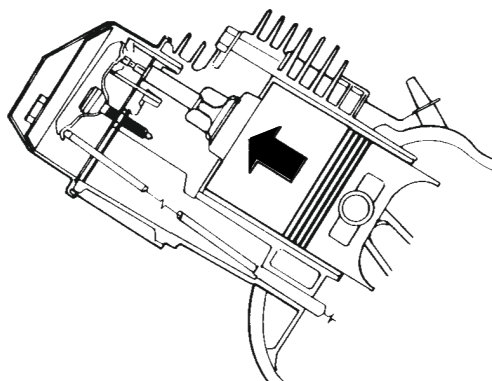
(A) Intake



001684

The piston is traveling from top dead center (TDC) to bottom dead center (BDC). The cam has opened the intake valve. The piston's downward movement in the cylinder creates a partial vacuum in the cylinder. Air at atmospheric pressure is drawn into the cylinder through the carburetor and is mixed with fuel in the carburetor. The fuel-air mixture flows through the open intake valve into the cylinder. The intake stroke is complete when the piston reaches BDC.

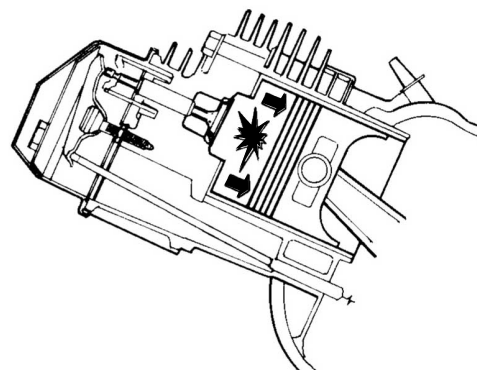
(B) Compression



001685

As the piston reaches BDC, both the intake and exhaust valves are closed. The piston moves upward toward TDC and the fuel-air mixture is compressed. Ignition occurs just before the piston reaches TDC.

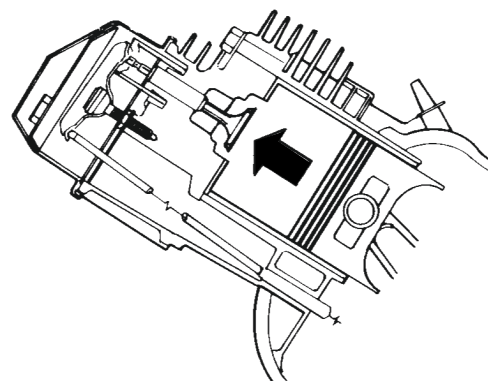
(C) Power



001686

Combustion is already in progress when the piston reaches TDC. The intake and exhaust valves remain closed as expanding gases of combustion force the piston downward.

(D) Exhaust



001687

The exhaust stroke begins when the piston has reached BDC and has started its upward movement. The intake valve is closed, and the exhaust valve is open to let gases escape.

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Section 3: General Information

Recommended Engine Oil

See the appropriate unit owner's manual.

Changing Engine Oil

WARNING

Skin irritation. Avoid prolonged or repeated contact with used motor oil. Used motor oil has been shown to cause skin cancer in laboratory animals. Thoroughly wash exposed areas with soap and water.

(000210)

See **Specifications** for crankcase oil capacities. Do not use special additives. Verify unit is level when filling with oil.

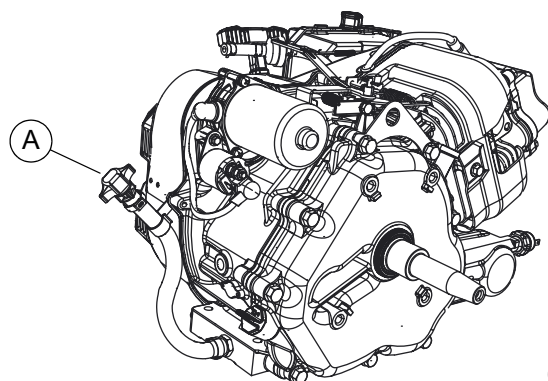
IMPORTANT NOTE: Do not overfill. Excessive oil will foam, smoke, and eventually damage internal parts of the engine.

Check and maintain oil level regularly. Change oil and filter after first eight hours of operation. Change oil and filter every 100 hours of operation thereafter. Change oil more often if engine is operated in dirty or dusty conditions, or if engine is operated under heavy loads or in high ambient air temperatures.

Procedure (Portable)

Proceed as follows to change engine oil on portable equipment:

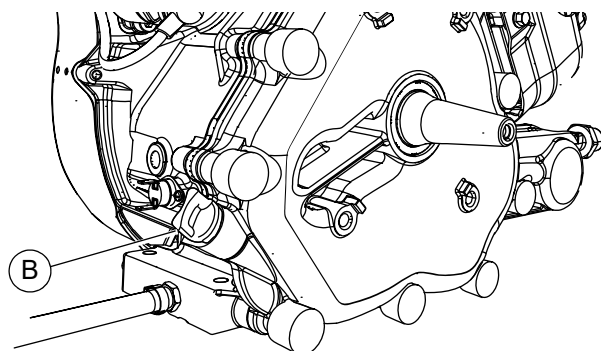
1. See **Figure 3-3**. Remove oil drain cap (A) and drain oil while engine is still warm. Install oil drain cap after oil has drained.



003904

Figure 3-1. Oil Drain (Portable)

2. See **Figure 3-2**. Remove oil fill cap (B) and refill slowly with new oil of required service classification and viscosity grade. Fill to bottom of threads in oil fill opening. Install oil fill cap.



001691

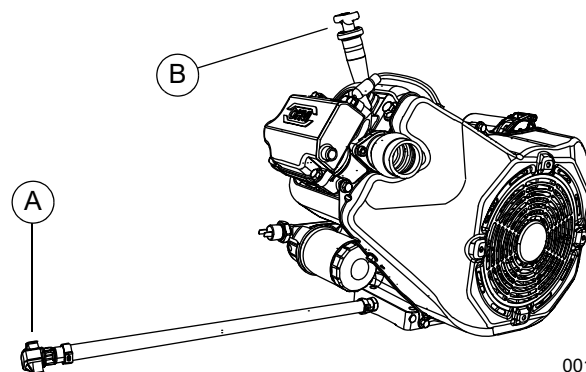
Figure 3-2. Oil Fill Cap

3. Start and run engine to check for oil leaks.

Procedure (HSB)

Proceed as follows to change engine oil on home standby units:

1. See **Figure 3-3**. Remove oil drain cap (A) and drain oil while engine is still warm. Install oil drain cap after oil has drained.



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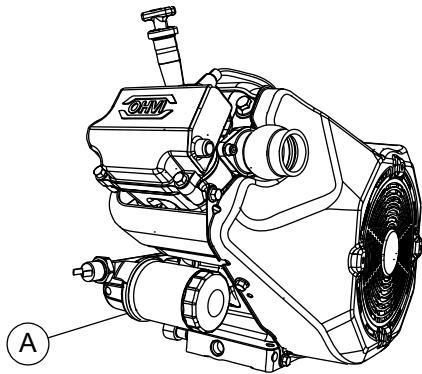
Figure 3-3. Oil Drain (HSB)

2. See **Figure 3-3**. Remove dipstick (B) and fill slowly with fresh oil of required service classification and viscosity grade. Fill to full mark on dipstick. When checking oil level, dipstick must be turned all the way in for accurate readings.
3. Start and run engine to inspect for oil leaks.

Changing the Oil Filter

Proceed as follows to change the oil filter:

1. See [Figure 3-4](#). For oil and oil filter change intervals, see engine maintenance requirements in the appropriate unit owner's manual.



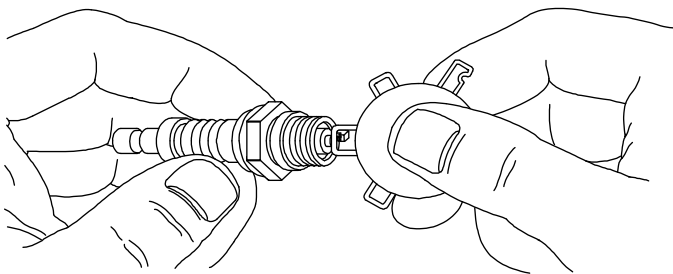
001692

Figure 3-4. Oil Filter

2. Lightly oil filter gasket with fresh clean engine oil before installing new filter.
3. Screw filter on by hand until gasket contacts filter adapter. Tighten three-quarters to one full turn farther.
4. Start and run engine at idle for 30 seconds and stop engine. Verify oil level and add if required.
5. Start engine and inspect for oil leaks.

Replacing Spark Plug

See engine specifications and maintenance requirements in the appropriate unit owner's manual. Replace spark plug if electrode is burned away, or if porcelain is cracked. Set spark plug gap to 0.020 in (0.50 mm). Torque spark plug to 18.4 ft-lbs (25 Nm).



000211

Figure 3-5. Checking Spark Plug Gap

See owner's manual for correct spark plug replacement.

Troubleshooting

Most complaints concerning engine operation can be classified as one or a combination of the following:

- Will not start
- Hard starting
- Lack of power
- Runs rough
- Vibration
- Overheating
- High oil consumption

NOTE: What appears to be an engine malfunction may be a fault of the powered equipment, rather than the engine. If equipment is suspect, see equipment affecting engine operation.

Systematic Inspection

If engine will not start and the cause of malfunction is not readily apparent, perform a systematic inspection in the following order:

1. Fuel
2. Ignition
3. Compression

This inspection, performed in a systematic manner, can usually be done in a matter of minutes. It is the quickest and surest method of determining the cause of failure. The basic inspection procedure is the same for all engine models. Any variation, by model, will be shown under the subject heading.

Checking Fuel

- Are the fuel tanks full?
- Is the fuel stale?
- Is the fuel tank vent open?
- Is the fuel shutoff valve open?

Checking Ignition

See [Ignition](#).

Checking Compression

Proceed as follows to check engine compression using master compression gauge kit (Generac part number GN04001):

1. Remove spark plug.
2. Insert an automotive type compression gauge into spark plug hole.
3. Crank engine until there is no further increase in pressure.

The highest reading obtained is the engine compression pressure.

- 426/460cc engines: Minimum 55 psi (379 kPa)

This minimum reading may appear low. This is due to the engine being equipped with a compression release mechanism to minimize starting effort. Actual running compression will be much higher (see [Compression Release](#)).

If compression is poor, look for one or more of the following causes:

- Loose cylinder head bolts
- Failed cylinder head gasket
- Burned / worn valves or valve seats
- Insufficient valve clearance
- Warped cylinder head
- Warped valve stem or worn valve guides
- Worn or broken piston ring(s)
- Worn or damaged cylinder bore
- Broken connecting rod

Cylinder Leakdown Test

A cylinder leakdown tester (Generac P/N 0F77000SRV) may be used to test the sealing capability of compression components of the cylinder at TDC, and quickly identify the problem component.

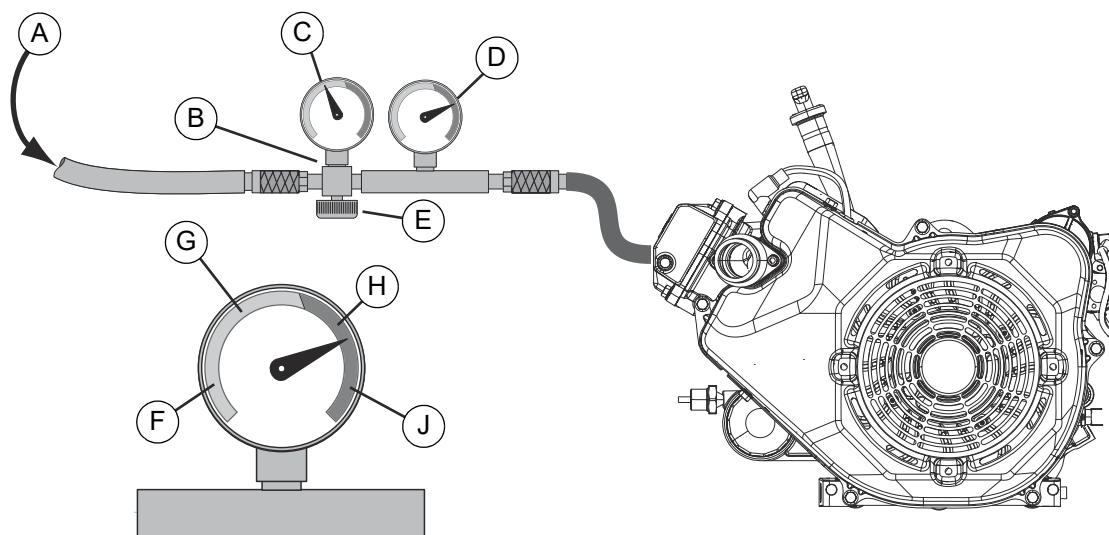
A cylinder leakdown test will assist in troubleshooting leaking valves or rings.

See [Figure 3-6](#). There are two gauges on the tester. One measures incoming air pressure, and the other measures percent of cylinder leakage.

Proceed as follows to perform a cylinder leakdown test:

1. Align piston at TDC.
2. Install cylinder leakdown tester in spark plug hole.
3. Pressurize cylinder to 90 psi (621 kPa).
4. Observe cylinder leakage and identify location of air leak.

NOTE: If leaking into intake or exhaust port, check valve clearance, valve face, and seat condition.



001694

Figure 3-6. Cylinder Leakdown Tester

A	Compressed air in
B	Air pressure regulator
C	Inlet gauge pressure set point
D	Outlet gauge pressure
E	Regular adjustment knob
F	Outlet gauge
G	Red range indicates unacceptable air leakage
H	Needle indicates minimal air leakage
J	Green range indicates acceptable air leakage

Results

- 0–10%—Cylinder leakage is acceptable
- 10–30%—Excessive cylinder leakage may exist
- 30–100%—Cylinder requires repair

Factors Affecting the Cylinder Condition

- Carburetion
- Crankcase vacuum
- Ignition timing
- A partially sheared flywheel key will affect ignition timing and engine performance
- Spark plug
 - A fouled spark plug may indicate that carburetor is out of adjustment
- Leak in spark plug wire
- Head gasket
- Intake manifold
- Valves
- Rings
- Piston
- Cylinder bore

Equipment Affecting Engine Operation

Frequently, what appears to be a problem with engine operation, such as hard starting, vibration, etc., may be the fault of equipment being powered rather than the engine itself. Listed are the most common effects of equipment problems—and what to look for as the most common cause.

Hard Starting or Will Not Start

- **Loose belt**—A loose belt, like a loose blade, can cause a backlash effect which will counteract engine cranking effort.
- **Starting under load**—Verify if unit is disengaged when engine is started; or, if engaged, should not have a heavy starting load.
- Inspect remote control assembly for correct adjustment.
- Inspect interlock system for shorted wires, loose or corroded connections, or faulty modules or switches.

Engine Won't Stop

- Inspect equipment ignition stop switch.
- Inspect for loose or disconnected equipment stop switch wire.
- Inspect ground wire harness on engine.
- See [Ignition](#) for test procedure.

Vibration

- Unit load out of balance—remove and balance.
- Mounting bolts loose—tighten.

Power Loss

- Bind or drag in unit. If possible, disengage engine and operate unit manually to feel for any binding action.
- Unit load has excess drag.

Noise

- Engine coupling or pulley. An oversize or worn coupling can result in knocking, usually under acceleration. Inspect for fit or tightness.
- Equipment needs lubrication.

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Section 4: Ignition

Major Components

See [Figure 4-1](#). Major components of the ignition system include engine run switch (if equipped), spark plug (A), ignition coil assembly (B), and engine flywheel (C).

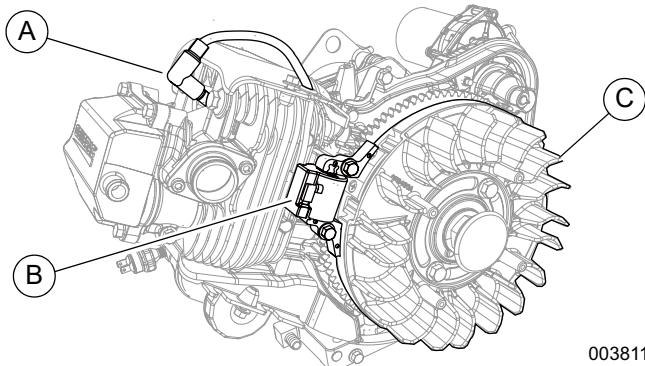


Figure 4-1. Ignition System Components

Ignition Coil Assembly

Solid-state components encapsulated in the ignition coil are not accessible and cannot be serviced. If the ignition coil is faulty, the entire assembly must be replaced. The air gap between ignition coil and flywheel magnet is fixed and nonadjustable.

See [Figure 4-2](#). Ignition coil assembly consists of spark plug boot (D), spark plug high tension lead (E), and ignition coil (F).

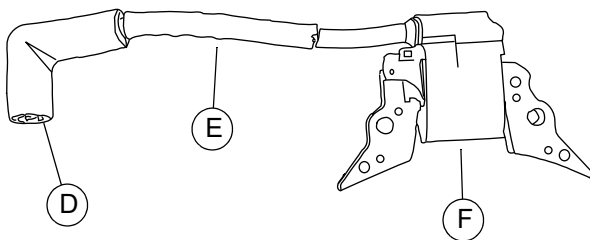


Figure 4-2. Ignition Coil Assembly

Spark Plug

The spark plug should be cleaned and re-gapped periodically. See specifications and maintenance requirements in the appropriate unit owner's manual.

See [Figure 4-3](#). Set spark plug gap to 0.020 in (0.50 mm).

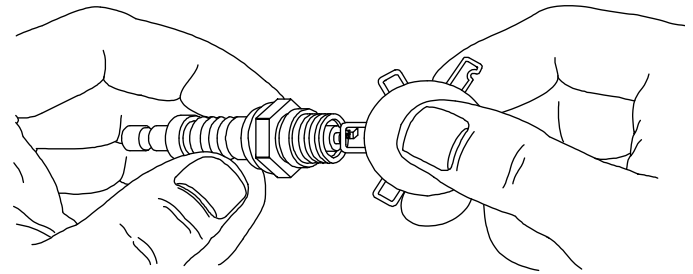


Figure 4-3. Setting Spark Plug Gap

NOTE: For correct spark plug replacement, see owner's manual for the specific product. Always use original Generac replacement parts.

Flywheel

Flywheel Key

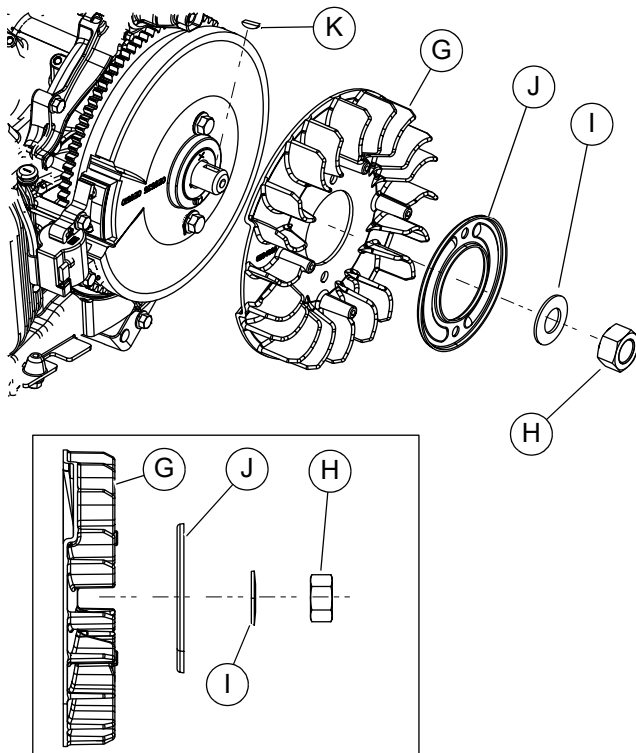
The flywheel taper is locked onto the crankshaft taper by the torque of the flywheel nut. A keyway is provided for alignment only and theoretically carries no load.

If the flywheel key becomes sheared, or even partially sheared, ignition timing can change. Incorrect timing can result in hard starting or failure to start.

Flywheel Removal

Proceed as follows to remove the flywheel:

1. See [Figure 4-4](#). Use a strap wrench to prevent the flywheel (G) from turning.



004541

Figure 4-4. Flywheel Removal

2. Remove flywheel nut (H). Then remove conical washer (I) and flywheel retainer (J).
3. Use a commercially available flywheel puller to remove flywheel from engine tapered shaft.

Flywheel Installation

Proceed as follows to install the flywheel:

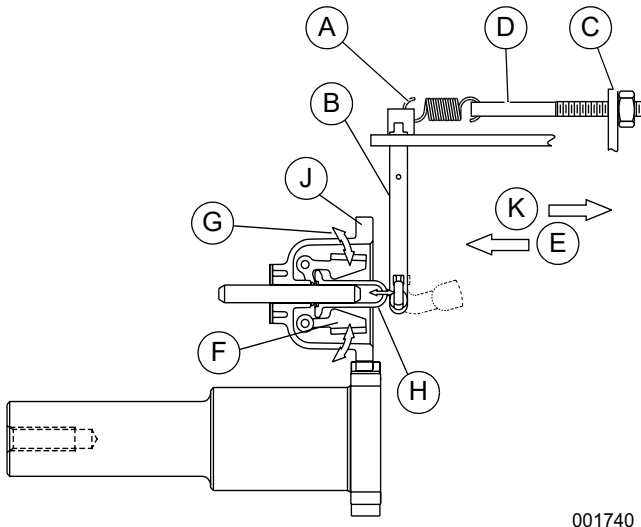
1. See [Figure 4-4](#). Align keyway in flywheel with woodruff key (K) on crankshaft.
2. Install flywheel, starter cup, and conical washer. Install flywheel nut and tighten to required torque.

The conical washer must be installed in the correct direction with the convex side facing out (away from flywheel).

Section 5: Mechanical Governor (If Equipped)

Governor Operation

See [Figure 5-1](#). The governor spring (A) tends to pull governor shaft (B), governor lever (C), and governor rod (D) toward the “Increase RPM” direction (E).



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A	Governor spring	F	Flyweight
B	Governor shaft	G	Centrifugal force
C	Governor lever	H	Governor spool
D	Governor rod	J	Governor gear
E	Increase RPM	K	Decrease RPM

Figure 5-1. Governor Operating Diagram

As engine speed increases, centripetal force acts on flyweight (F). When flyweight centrifugal force (G) exceeds governor spring force, governor spool (H) is moved axially by flyweight force. This causes governor gear (J) to rotate, which then moves governor lever and governor rod toward the “Decrease RPM” direction (K).

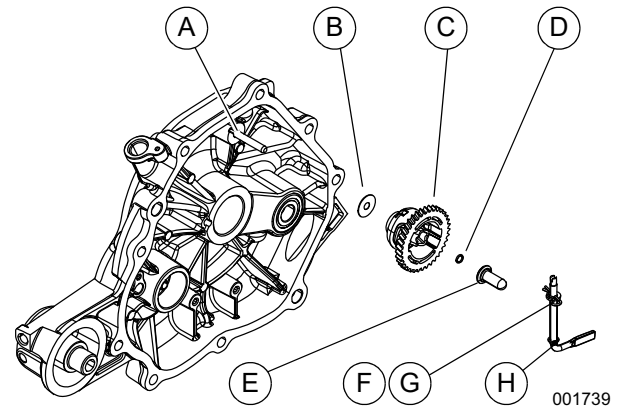
As engine speed decreases, a point is reached where governor spring force is greater than flyweight centrifugal force. Governor spring force then moves governor lever and governor rod toward “Increase RPM.”

Governor action consists of a series of small rpm overshoots and undershoots, as governor spring force increases rpm and flyweight centripetal force decreases rpm.

The point at which spring tension and flyweight force are equal is the governed speed of the engine.

Governor Internal Parts

See [Figure 5-2](#). The governor gear shaft (A) is pressed into a bore in the engine crankcase cover. Internal governor components include a thrust washer (B), governor gear assembly (C), snap ring (D), governor spool (E), and governor arm (H). A lock pin and washer (F, G) retain the governor arm in the crankcase cover.



001739

A	Governor gear shaft	E	Governor spool
B	Thrust washer	F	Lock pin
C	Governor gear assembly	G	Washer
D	Snap ring	H	Governor arm

Figure 5-2. Internal Governor Parts

The governor gear assembly houses governor flyweights.

Flyweight movement causes governor spool to move axially. In turn, axial movement of governor spool results in rotation of the governor arm.

Governor Assembly

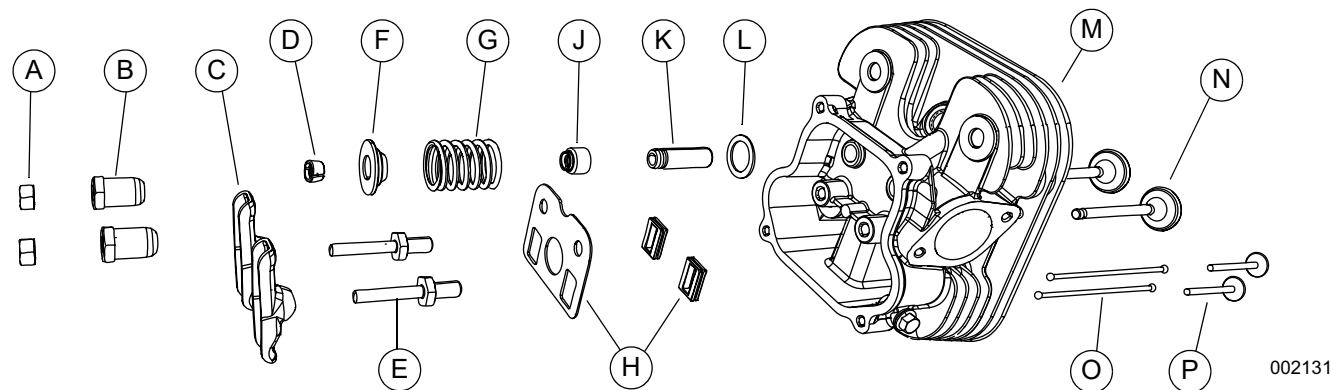
See [Assembling Internal Governor \(Portable\)](#).

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Section 6: Cylinder Head and Valves

Major Components

Valve train components are shown in [Figure 6-1](#) below.



ID	Qty.	Description	ID	Qty.	Description
A	2	Jam nut	J	2	Valve stem seal
B	2	Rocker adjusting ball	K	2	Valve guide (not removable)
C	2	Rocker arm	L	2	Valve spring washer
D	2	Valve spring keeper	M	1	Cylinder head
E	2	Rocker stud	N	2	Valve
F	2	Valve spring retainer	O	2	Pushrod
G	4	Valve spring	P	2	Tappet
H	1	Pushrod guide plate			

Figure 6-1. Valve Train Components

Disassembly

Proceed as follows to disassemble the cylinder head:

1. See [Figure 6-2](#). Remove blower housing (A) and upper and lower wrappers (B).

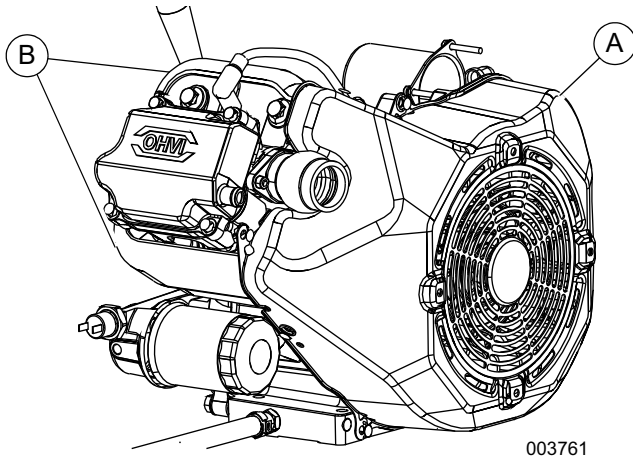


Figure 6-2. Blower Housing and Wrappers

2. See [Figure 6-3](#). Remove oil breather tube. Remove the four M6-1.00 x 12 mm screws and the rocker arm cover.

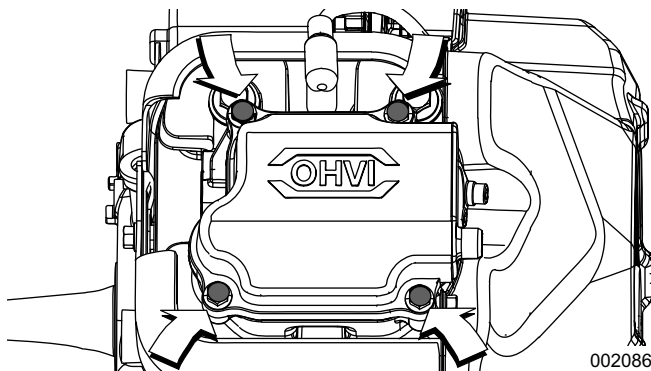


Figure 6-3. Rocker Arm Cover Removal

NOTE: Whenever the rocker arm cover is removed, the rocker arm cover gasket must be replaced to ensure a correct seal. See [Gasket Removal](#) for more information.

3. See [Figure 6-4](#). Loosen rocker arm jam nuts (C) on the pivot ball studs. Loosen pivot ball studs. Remove the two pivot ball studs, rocker arms, and jam nuts.
4. Remove push rod guide plate.

NOTE: Always keep intake and exhaust valve parts separated. Intake and exhaust pivot ball studs, rocker arms, and push rods are identical. Wear patterns will be different.

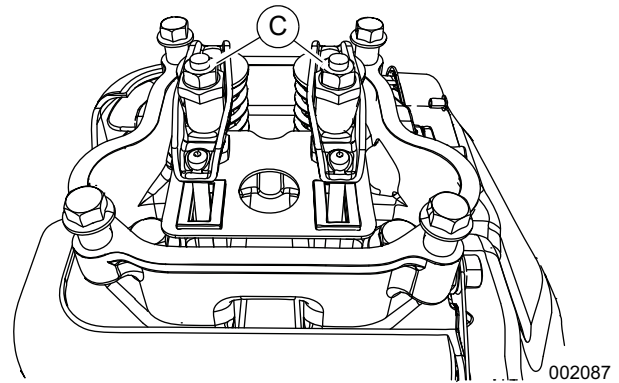


Figure 6-4. Rocker Arms, Pivot Ball Studs, Push Rods, and Push Rod Plate

5. Remove push rods.
6. See [Figure 6-5](#). Remove cylinder head bolts, then the cylinder head and cylinder head gasket.

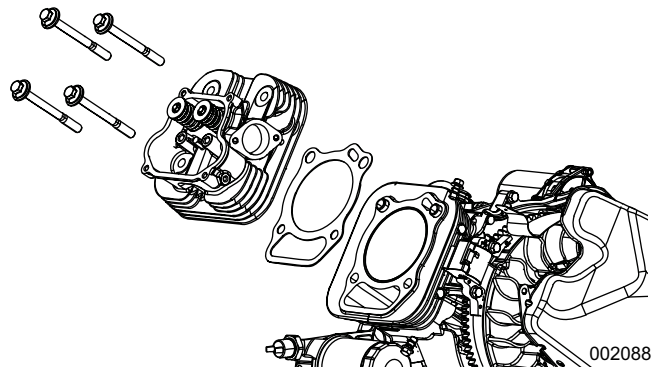


Figure 6-5. Cylinder Head Removal

NOTE: Replace the cylinder head gasket every time the cylinder head is removed. The new cylinder head gasket must be free of nicks and scratches, as these could cause leaks.

Gasket Removal

Verify sealing surface is completely clean and free of old gasket material before installing a new gasket. Residual gasket material or dirt can prevent the new gasket from sealing completely. A poorly sealed gasket may cause the engine to leak oil, lose compression, or overheat.

It is very important to completely remove old gasket material without damaging sealing surfaces. Aluminum is very soft and easy to scratch and gouge. If the underlying surface is damaged, even a new gasket may not seal completely. Generac does not recommend using a wire brush or abrasive sanding equipment to remove gaskets. Use a metal gasket scraper specifically designed for this purpose. Scrape gasket at a shallow angle—almost parallel to the sealing surface. Wipe sealing surface with a clean, dry, lint-free rag after removing gasket residue.

If the gasket is difficult to scrape off, spray it with an aerosol gasket removal compound. Allow compound to soak into the gasket for the length of time recommended by the manufacturer (typically 10–20 minutes). Then use the gasket scraper again.

NOTE: Some gasket removal compounds contain chemicals that can damage plastics. Read and follow directions on the product label before use.

Valve Spring Keepers Removal



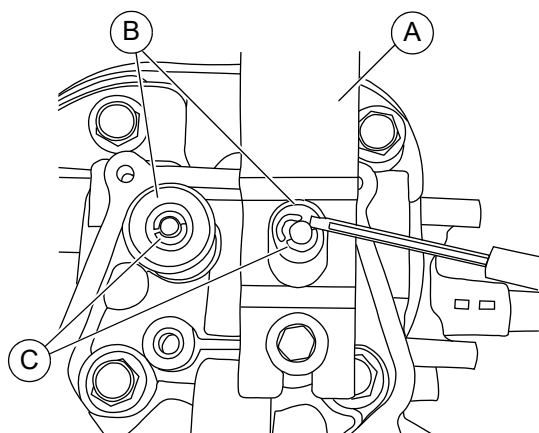
⚠ WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes.

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Proceed as follows to remove valve spring keepers:

1. See **Figure 6-6**. Place a commercially available small engine valve spring compressor (A) squarely on top of the valve spring retainer (B). The split valve keepers (C) should "pop" out.



002089

Figure 6-6. Remove Valve Spring Keepers

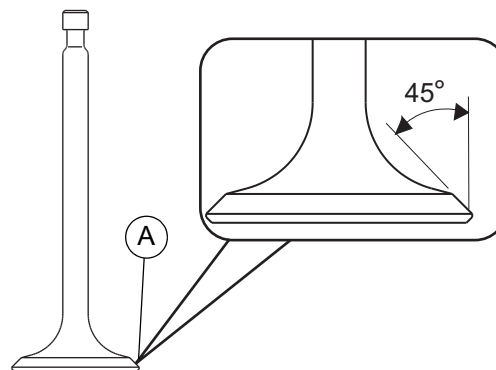
2. With split valve spring keepers removed, remove valve spring retainers, valve springs, valve spring washers, and intake and exhaust valves.
3. Clean all parts. Remove carbon from valve heads and stems. Inspect valves and valve seats. Service parts as outlined under **Valve Service**.

Valve Service

Replace valves if they are damaged, distorted, or if the margin is ground to less than 0.039 in (1.0 mm). If the valve is in usable condition, grind and lap the valve as described in the procedures below.

Grinding and Lapping the Valve Face

See **Figure 6-7**. Use a valve grinder or valve refacer to grind the valve face (A) to a 45° angle.



002090

Figure 6-7. Typical Seat

After reconditioning, lap the valves with a suitable valve lapping tool and valve lapping compound.

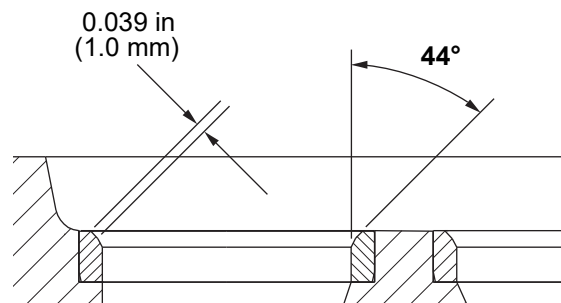
NOTE: Correct lapping of valves and seats will remove grinding marks and ensure a good seal between the valve and seat. After lapping, remove all lapping compound from the valve faces and seats.

If using a commercially available valve and seat system, it is not required to lap the valve or seat.

Valve Seats

Valve seats are not replaceable. If they are burned or pitted, they can be reground using a grinding stone or valve reseater.

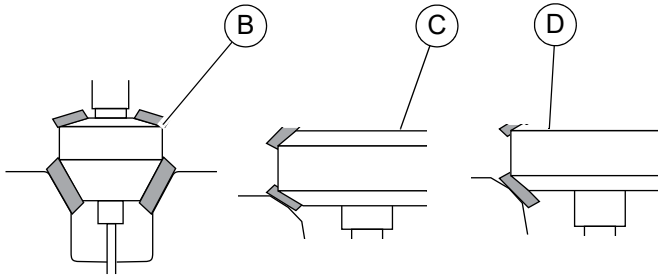
See **Figure 6-8**. Grind seats at a 44° angle and to a width of 0.039 in (1.0 mm) on intake and 0.059 in (1.5 mm) on exhaust.



002091

Figure 6-8. Valve Seat

See [Figure 6-9](#). Commercially available valve cutting systems use cutters of three different degrees to produce a 44° angle.



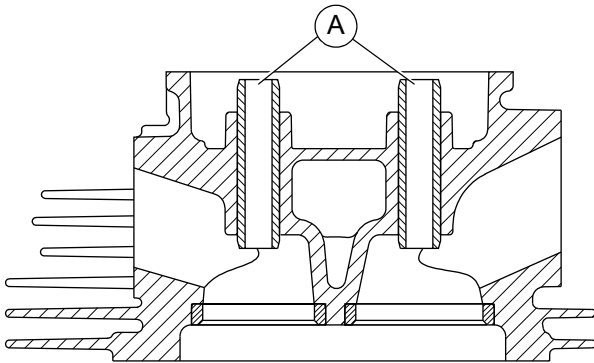
002092

Figure 6-9. Three-Step Method: Cutting a Valve Seat

1. Use a 60° cutter (B) to clean and narrow the seat from the bottom toward the center.
2. Use a 31° cutter (C) to clean and narrow the seat from the top to the center.
3. Use a 44° cutter (D) to cut in the seat to a width of 0.039 in (1.0 mm) on intake and 0.059 in (1.5 mm) on exhaust.

Valve Guides

See [Figure 6-10](#). Valve guides (A) are permanently installed in the cylinder head and cannot be replaced.



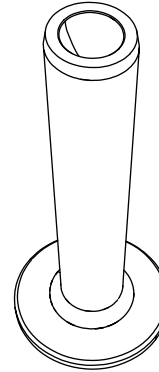
002093

Figure 6-10. Valve Guides

At the time of this writing, oversize valves are NOT available. Thus, if valve guides are worn or damaged, the guides cannot be reamed to accommodate an oversize valve stem.

Valve Tappets

See [Figure 6-11](#). The valve tappets are removed when the camshaft is removed. Intake and exhaust valve tappets are identical. However, once a wear pattern has been established, the two parts should not be interchanged.

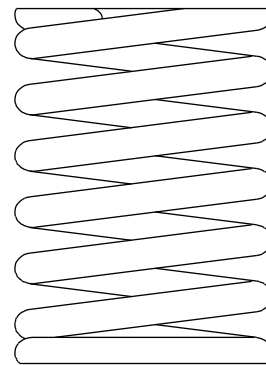


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Figure 6-11. Valve Tappet

Valve Springs

See [Figure 6-12](#). Inspect valve springs. Measure length of uncompressed spring (known as “free length”). Measure spring length when a known load is applied to the spring. See [Specifications](#).



002095

Figure 6-12. Valve Spring

Installation

After the valve train components have been correctly inspected and serviced, proceed as follows to install the components:

1. See [Figure 6-14](#). Install intake (B) and exhaust (A) valves through the correct valve guides in the cylinder head.

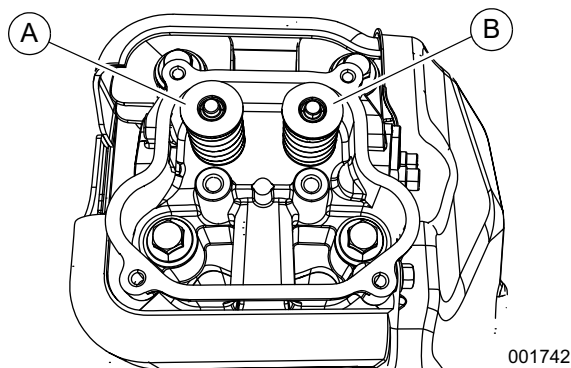


Figure 6-13. Intake and Exhaust Valves

- a. The exhaust valve has the smaller head.
- b. The intake valve has the larger head.
- c. Valve seat sizes will match their respective valve head sizes.
- d. The exhaust valve stem is smaller than the intake valve stem.

IMPORTANT NOTE: Lubricate valve stems and valve guides before assembly.

2. If intake valve has a valve stem seal:
 - a. Install intake valve stem seal.
 - b. Install valve spring washer, valve spring, and valve spring retainer.
 - c. Install valve spring keepers.

NOTE: To install valve spring keepers, use valve spring compressor to compress valve spring enough to insert one keeper half. Compress valve spring again to insert the other keeper half.

3. If exhaust valve does not have a valve stem seal:
 - a. Install valve spring washer, valve spring, and valve spring retainer.
 - b. Install valve spring keepers.
4. After both valves have been correctly installed in the head, position new head gasket and install cylinder head.

NOTE: Head gasket is coated with a special substance for better sealing. It must be free of nicks, scratches, and other faults, or leakage could result.

5. See [Figure 6-14](#). Install cylinder head bolts and tighten them in the sequence shown.

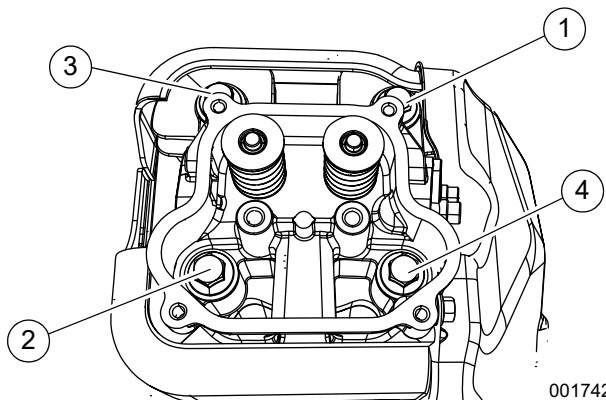


Figure 6-14. Head Bolt Tightening Sequence

6. See [Figure 6-15](#). Position push rod guide plate on the head. Then install the rocker arms and pivot ball studs. Install the jam nuts far enough to hold the guide plate in position. Valve clearance will be adjusted later.

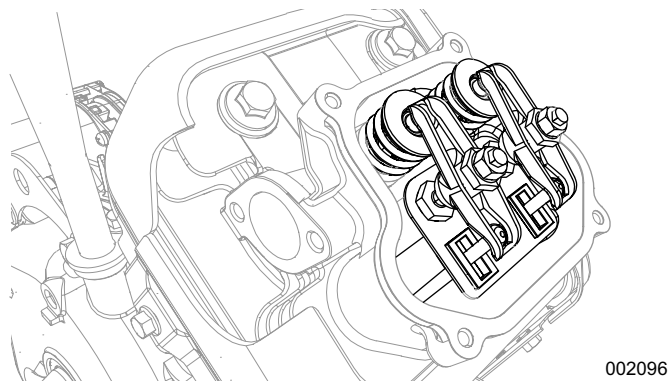


Figure 6-15. Install Rocker Arm and Pivot Ball Stud

7. See [Figure 6-16](#). Rotate rocker arm as shown to install push rod. Insert push rod through push rod guide plate, with either end resting against the tappet. Place rocker arm over push rod. Alignment is correct when push rod ball rests in rocker arm socket.

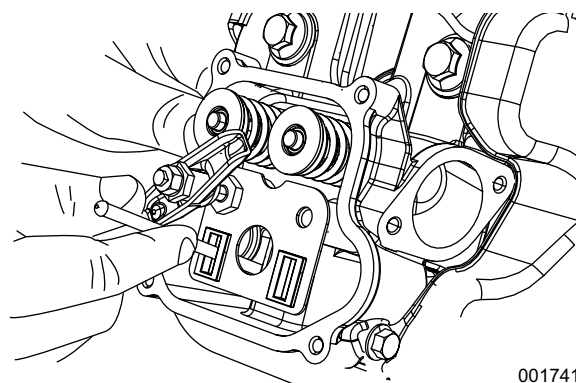


Figure 6-16. Install Push Rods

Adjusting Valve Clearance

Adjust valve clearance with engine at room temperature. The piston should be at TDC of its compression stroke (both valves closed). Proceed as follows to adjust valve clearance:

1. See **Figure 6-17**. Loosen rocker arm jam nut. Use a box wrench to turn pivot ball stud while inspecting the clearance between rocker arm and valve stem with a feeler gauge (A). See **Specifications** for correct clearance.

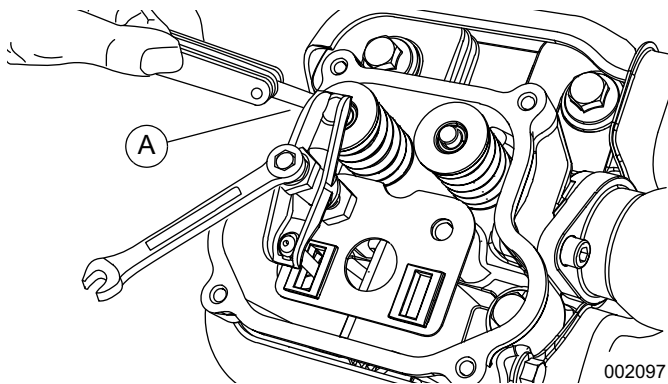


Figure 6-17. Adjusting Valve Clearance

2. See **Figure 6-18**. When clearance is correct, hold pivot ball stud with box wrench and tighten rocker arm jam nut to the specified torque with an open-ended wrench. Verify valve clearance did not change after tightening jam nut.

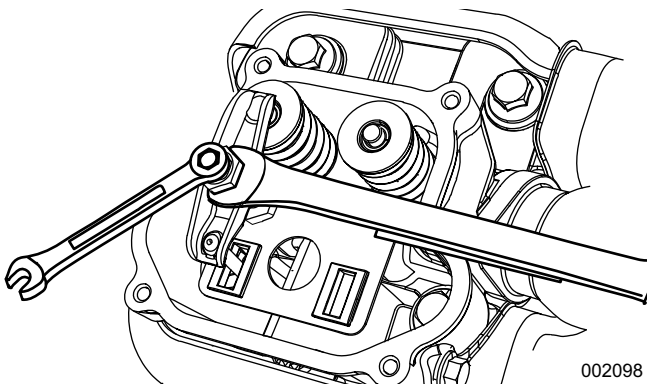


Figure 6-18. Tightening Rocker Arm Jam Nut

Installing Rocker Arm Cover

Proceed as follows to install the rocker arm cover:

1. See **Figure 6-19**. Using a new rocker arm cover gasket (A), install rocker arm cover (B), and retain with four screws (C).

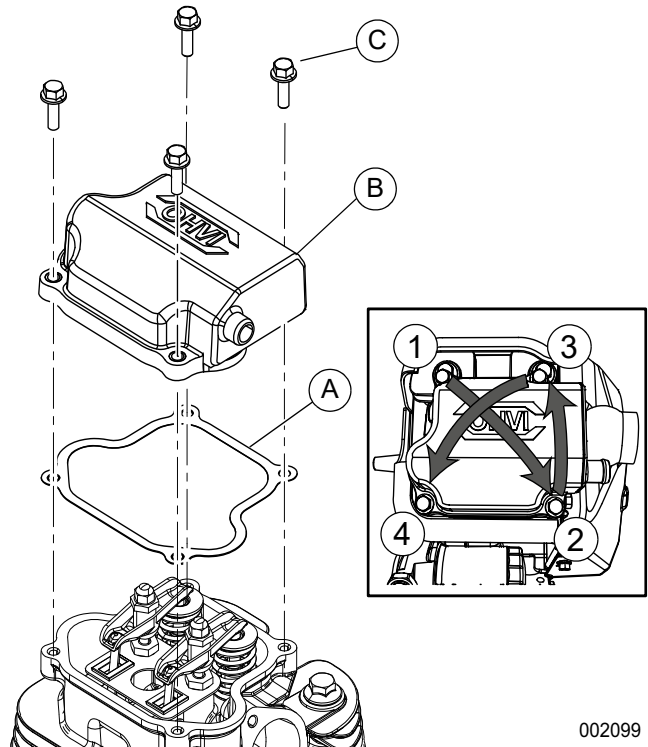


Figure 6-19. Install Rocker Arm Cover

2. Tighten each screw to 20 **in-lbs** (2.24 Nm), working diagonally from corner to corner.
3. Repeat the same pattern, this time tightening each screw to 48 **in-lbs** (5.4 Nm). DO NOT EXCEED this torque value.
4. Install oil breather tube.

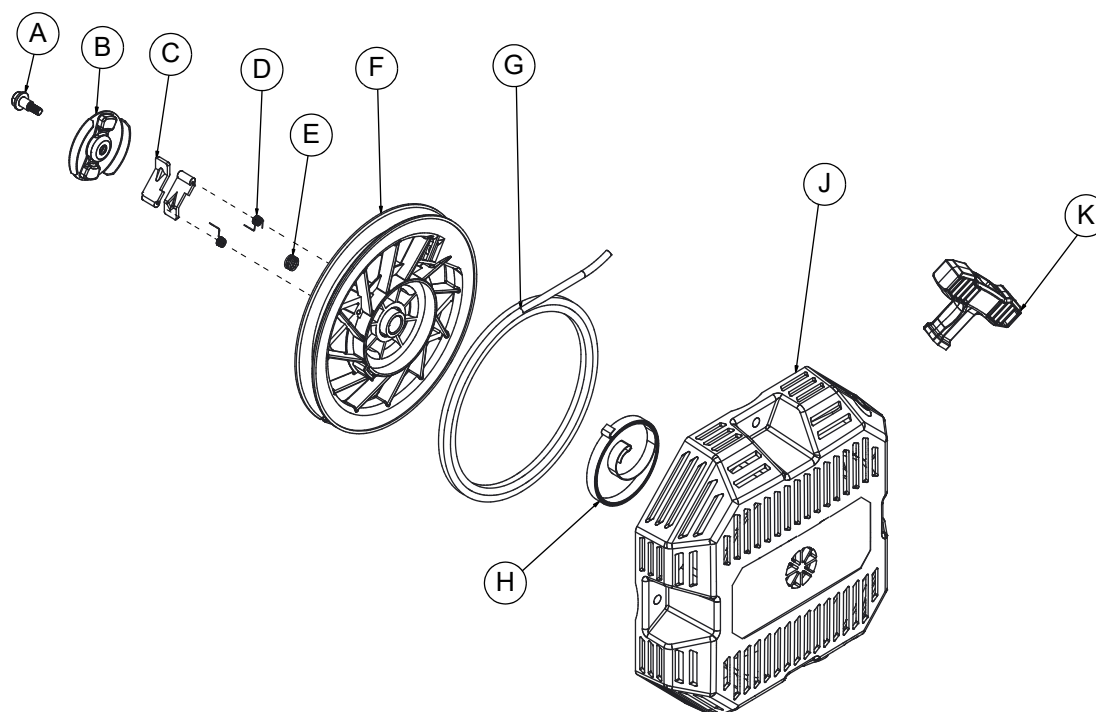
Section 7: Recoil Starter (If Equipped)

Overview

The recoil starter used on portable equipment engines is a manual starter that uses a spring to rewind the starter rope after it has been pulled. Pulling the rope winds a clock-type spiral spring in the starter housing. When the rope is released, the spring unwinds and causes the rope to wind around the pulley.

When the rope is pulled outward and away from the engine, spring-loaded “starter dogs” engage the pulley, and the engine is cranked.

When the rope is allowed to return, the starter dogs disengage from the pulley.



001730

A	Screw	F	Pulley
B	Retainer pawl	G	Recoil rope
C	Recoil arm	H	Torsional spring
D	Dog spring	J	Recoil cover
E	Spring, axis recoil	K	Recoil handle

Figure 7-1. Exploded View of Recoil Starter

Recoil Assembly Removal

See [Figure 7-2](#). The recoil starter assembly is fastened to the engine blower housing with screws and lock washers. Remove the screws and lock washers, then remove the complete starter rewind assembly.

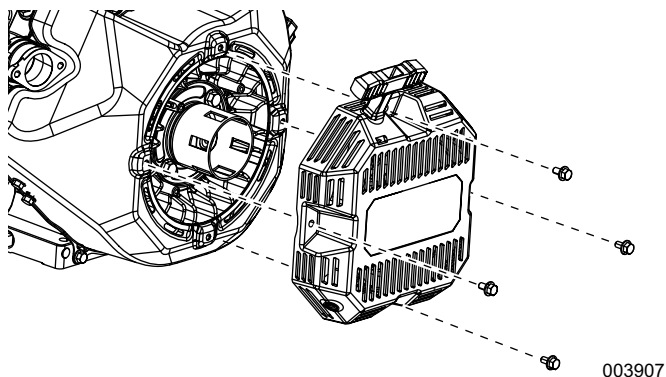


Figure 7-2. Recoil Assembly Removal

Disassembly and Repair

Disassembly and repair of the recoil starter is not cost-effective. Replace the entire recoil assembly.

Section 8: Electric Starter (If Equipped)

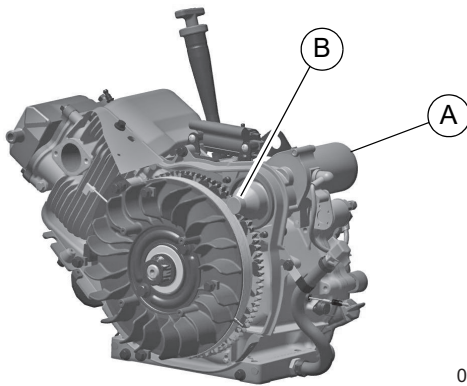
Introduction

The Generac 426/460cc engine may be equipped with a 12 volt DC electric cranking system. Such a system converts electrical energy from a battery into mechanical energy at the starter motor. The starter motor then turns the engine over.

NOTE: Cranking systems discussed in this manual are typical systems. Actual cranking system used in specific applications may differ. See wiring diagram and/or electrical schematic in owner's manual for specific applications.

Starter Motor Operating Principles

See [Figure 8-1](#). Closing the circuit to the starter motor (A) allows battery current to flow through a commutator, to the loops of wire in the armature, and back to the battery. The interaction of the magnetic fields causes the armature to revolve.



009677

Figure 8-1. Electric Starter

The armature rotates at a relatively high speed to provide sufficient torque to crank the engine. The required engine cranking speed is relatively slow, so the starter motor is equipped with a small drive pinion that meshes with the teeth of a flywheel ring gear (B) to crank engine. The large ring gear and small starter pinion gear results in a gear reduction that can vary in ratio up to 19-to-1. This reduction allows the starter to rotate at high speeds while cranking the engine at low speeds.

Speed increases quickly when the engine starts. For example, if the engine reaches 100 rpm and starter pinion remains meshed with ring gear, the starter armature would spin at about 1,900 rpm (19-to-1 ratio). To prevent damage to the armature, a "Bendix" drive mechanism is used to disengage starter pinion from the ring gear when the engine has started.

Bendix Drive

The armature starts to rotate when field coils of the starter drive are energized. A loose fitting sleeve inside the pinion gear rotates with the armature. This sleeve has large spiral threads on its surface matching the pinion gear's internal threads. The sleeve rotates with the armature, and pinion gear rotates on the threads to move outward on the sleeve. Outward movement of the pinion gear causes the pinion gear to mesh with the flywheel ring gear. As soon as the pinion gear engages a stop on the sleeve, the pinion gear rotates the ring gear and engine.

The engine rotates faster than the armature on startup. This causes the pinion to spin back on the spiral threads of the sleeve and out of engagement with the ring gear.

Starter Motor Repair

A faulty starter motor should be removed and replaced. Starter motor disassembly and repair is not cost-effective.

Removing the Starter Motor



Accidental start-up. Disconnect negative battery terminal before servicing to prevent accidental engine rotation. Failure to do so could result in death or serious injury.

(000148a)

To remove starter motor from engine, loosen and disconnect the starter wire, then remove the two hex head screws holding the starter in place.

Installing the Starter Motor

Reverse previous steps to install starter motor.

Testing the Starter Motor

Checking the Pinion

The pinion gear should rise and engage the ring gear when the starter motor is activated. If the pinion gear does not rise normally, inspect the large spiral threads of the sleeve and pinion for binding or sticking.

Tools for Starter Performance Test

The following equipment may be used to complete a performance test of the starter motor:

- a digital multimeter (DMM)
- a tachometer capable of reading up to 1,500 rpm
- a fully charged 12 volt battery

Starter Performance Test

Proceed as follows to test starter performance:

1. Set voltmeter (A) to read DC amps.
2. See **Figure 8-2**. Connect voltmeter as shown.

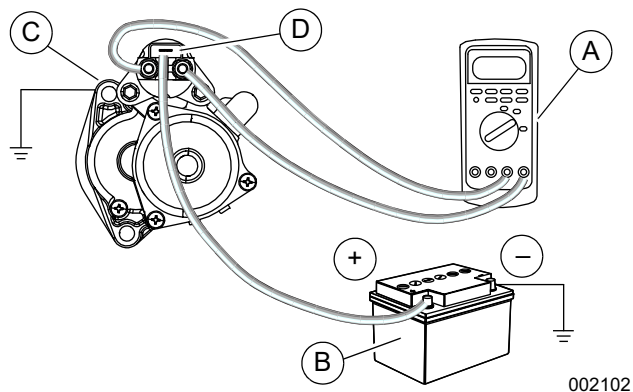


Figure 8-2. Starter Performance Test

3. Ground battery (B) and starter casting (C).
4. Connect one end of jumper cable to starter spade terminal (D).
5. Touch free end of jumper cable to positive battery lead to close the solenoid. An audible click should be heard as the solenoid switch “closes.”

Starter Motor Performance Specifications at 12 volts DC:

- Minimum motor rpm = 800
- Maximum amps = 9

Replace starter motor if it does not perform satisfactorily.

Section 9: Lubrication System

Overview

The engine lubrication system serves to (a) reduce friction between moving parts, (b) cool the engine, and (c) establish a negative pressure in the crankcase to prevent oil seal leakage.

Major components include the following:

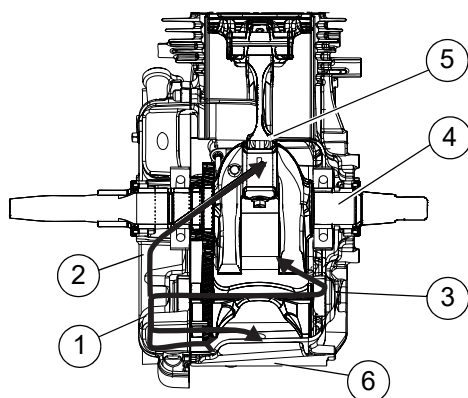
- Oil pickup assembly
- Oil pump
- Crankshaft oil seals
- Pressure relief valve
- Breather assembly
- Crankcase cover
- Low Oil Pressure Switch

Optional Systems

In addition to the major components listed above, some engines may be equipped with an optional oil filter.

Oil Flow

See [Figure 9-1](#). An oil pickup screen extends into the crankcase. The oil pump draws crankcase oil through the oil pickup screen and delivers it to areas requiring lubrication.



004040

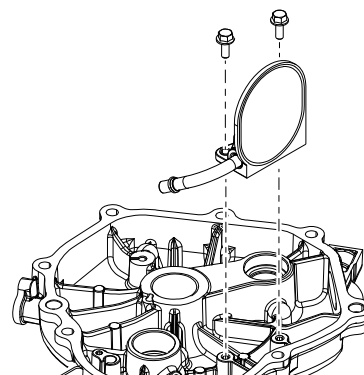
Figure 9-1. Lubrication System Diagram

1. If equipped, oil will pass through an oil filter and through a cored passage in the crankcase cover.
2. The cored passage in the crankcase cover allows oil to reach the oil pump and one camshaft bearing surface. Oil flows through the cored passage in the cover and also flows to one crankshaft bearing.
3. Oil flows through the hollow camshaft to lubricate the camshaft bearing at the opposite end of the camshaft.
4. Oil flows through the crankshaft and to the crank throw to lubricate the rod bearing.

5. If oil pressure exceeds 40 psi (276 kPa), the oil pressure relief valve opens.

Oil Pickup Assembly

See [Figure 9-2](#). The oil pickup assembly consists of a cylindrical plastic tube with a screen. Two o-ring seals are installed in grooves at one end of the cylinder. The pickup slides into a bore of the oil filter pad in the crankcase cover and extends into the interior of the crankcase. Once installed in the crankcase cover bore, the pickup is retained by either of the following methods:



001743

Figure 9-2. Oil Pickup Assembly

Inspection

Remove o-ring seals, then clean pickup in solvent. Inspect plastic body for cracks or other damage. Inspect pickup screen for damage or clogging. Verify hollow tube is free of obstructions.

Install two o-ring seals prior to reassembly. Replace oil pickup assembly if it is damaged, if pickup screen is torn or plugged, or if hollow tube is plugged. Replace damaged o-rings.

Oil Pump

Description

See [Figure 9-3](#). The oil pump is a trochoid type. The inner rotor (A) rotates on a shaft provided in the camshaft bore of the crankcase cover. The outer rotor (B) is installed over two drive pins on the end of the camshaft (C) and is driven by camshaft rotation.

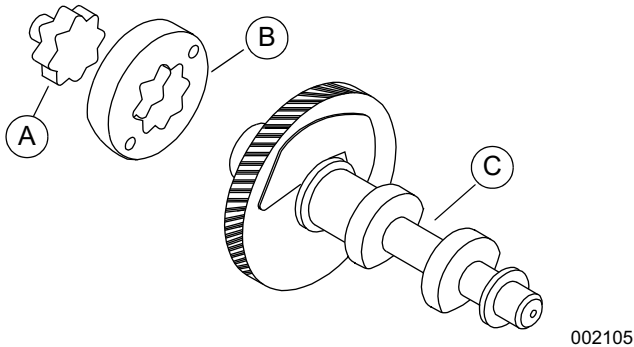


Figure 9-3. Oil Pump

Inspection

Proceed as follows to inspect the oil pump:

1. See [Figure 9-4](#). Inspect the inner and outer rotors for damage and wear.

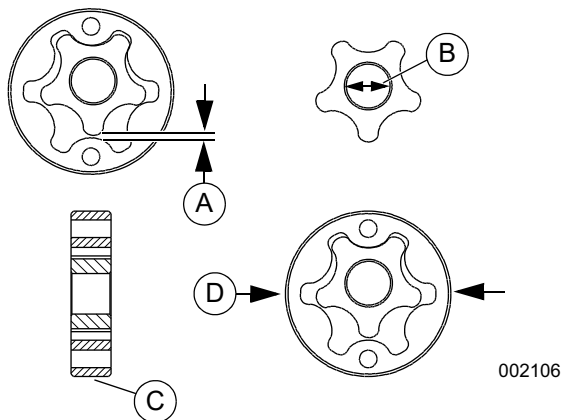


Figure 9-4. Inner and Outer Rotor Check Points

2. Use a feeler gauge to check tip clearance of inner rotor (A), with rotor installed on shaft in crankcase cover. Tip clearance is 0.015 in (0.38 mm) maximum.
3. Measure bore inner diameter (B) of the inner rotor. Bore inner diameter is 0.357 in (9.068 mm) maximum.
4. Measure inner rotor thickness (C). Inner rotor thickness is 0.476 in (12.09 mm) maximum.
5. Measure outer rotor outside diameter (D). Outer rotor outside diameter is 1.2909 in (32.789 mm) maximum.

6. Inspect outer rotor drive pins on the camshaft for breaking, bending, or other damage. The two roll pins can be easily removed and replaced.

Replace any oil pump component that is damaged or worn excessively.

NOTE: The shaft on which the inner oil pump rotor rotates is NOT replaceable. Crankcase cover must be replaced if the shaft is damaged or worn.

Pressure Relief Valve

Description

A ball type pressure relief valve is located in a bore of the crankcase cover. The valve serves to limit oil pressure to a maximum value. The ball remains seated as long as oil pressure is below approximately 40 psi (276 kPa). If oil pressure increases above that value, the ball is forced off its seat to relieve excess oil pressure into the crankcase. The ball and spring are held in place by a retainer.

Inspection

Proceed as follows to inspect the pressure relief valve:

1. See [Figure 9-5](#). Remove screw (A) that fastens retainer (B) to crankcase. Remove retainer, oil pressure relief valve spring (C), and ball (D). Clean all parts in solvent.

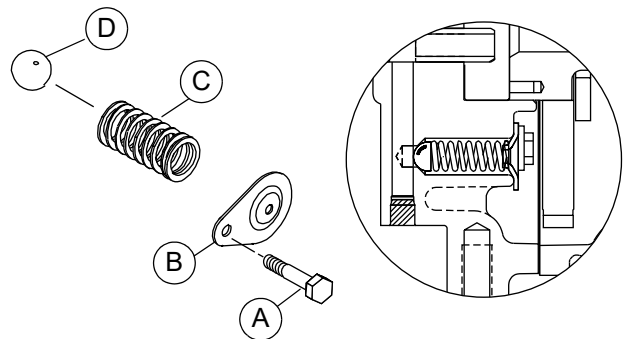


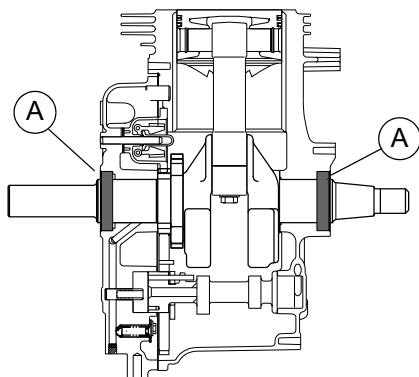
Figure 9-5. Oil Pressure Relief Valve

2. Inspect ball and retainer for damage and wear. Replace any damaged or worn part. Inspect spring; replace if damaged or worn.
3. Measure free length of the oil pressure relief valve spring. See [Specifications](#) for correct oil pressure relief valve spring dimensions. Replace oil pressure relief valve spring if it is not the correct length.

Crankshaft Oil Seals

Description

See [Figure 9-6](#). An oil seal (A) is provided in the crankcase cover and in the crankcase to prevent oil leakage past crankshaft journals.



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Figure 9-6. Crankshaft Oil Seals

Seal Replacement

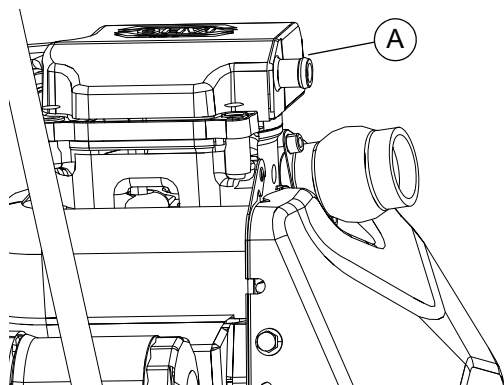
Proceed as follows to replace a faulty oil seal:

1. If the crankshaft has been removed, old seals can be removed by tapping them out with a screwdriver or by tapping them out from the inside.
2. If the crankshaft is installed, use a commercially available oil seal puller to remove the seals.
3. Always use a seal protector when installing the crankshaft into its bearing bore, and when installing the crankcase cover over the crankshaft.

Breather Assembly

Description

See [Figure 9-7](#). A crankcase breather (A) is located in the rocker arm cover. The breather serves to maintain a net negative pressure in the engine crankcase and to prevent oil from being forced past the oil seals, gaskets, or piston rings.



003804

Figure 9-7. Crankcase Breather

The check valve allows excess pressure to be vented out of the crankcase and to the air inlet side of the carburetor through the breather tube. Drain holes allow condensed oil vapors to return to the crankcase.

NOTE: The breather assembly on this engine is ultrasonically installed. There are no serviceable parts.

Crankcase Cover

Description



Personal injury. Wear appropriate personal protective equipment at all times while operating and servicing unit. Failure to do so could result in personal injury.

(000419)

The die-cast aluminum crankcase cover is fastened to the crankcase with flanged head bolts.

Install a new gasket between crankcase cover and crankcase each time crankcase cover is removed. Machined bores are provided in crankcase cover for (a) oil pump rotors and camshaft, (b) crankshaft, (c) balancer, (d) governor gear assembly, and (e) oil pickup assembly. Cored oil passages are provided from the pickup bore to the pump and from the pump to the crankshaft bore.

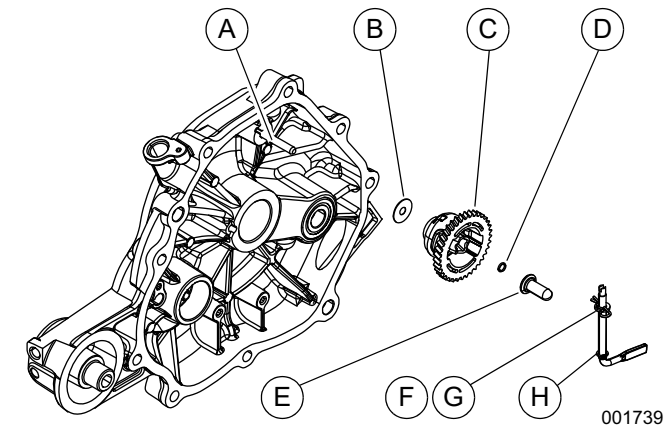
Inspection

Clean crankcase cover and blow dry with compressed air. Use compressed air to blow out all bores and oil passages. Inspect crankcase cover for cracks, damage, etc. Verify crankshaft, camshaft, and balancer bearing bore diameters as outlined in [Specifications](#).

Governor Gear Installation

Proceed as follows to install governor gear:

- 1. See [Figure 9-8](#). Apply engine oil to the governor gear shaft.



A	Governor gear shaft
B	Thrust washer
C	Governor gear assembly
D	Snap ring
E	Governor spool
F	Lock pin
G	Washer
H	Governor arm

Figure 9-8. Governor Assembly

- 2. Install governor gear thrust washer over the governor gear shaft followed by the governor gear assembly.
- 3. Slide governor snap ring to a distance of 0.31 in (8 mm) down over the governor gear shaft.
- 4. Lift gear assembly up to governor snap ring.
- 5. Install governor spool over governor gear shaft so that the flange is under the flyweights.
- 6. Slide governor gear assembly, snap ring, and governor spool down governor shaft until bottomed out.

Crankcase Cover Installation

Proceed as follows to install crankcase cover:

- 1. Rest engine on its side with crankcase cover flange facing upward.
- 2. Install valve tappets, crankshaft, connecting rod assembly, camshaft, and balancer as described in [Cylinder Head and Valves](#).
- 3. Verify timing marks on the crankshaft, camshaft, and balancer gears are correctly aligned.

- 4. See [Figure 9-9](#). Place crankcase cover gasket (A) onto crankcase flange.

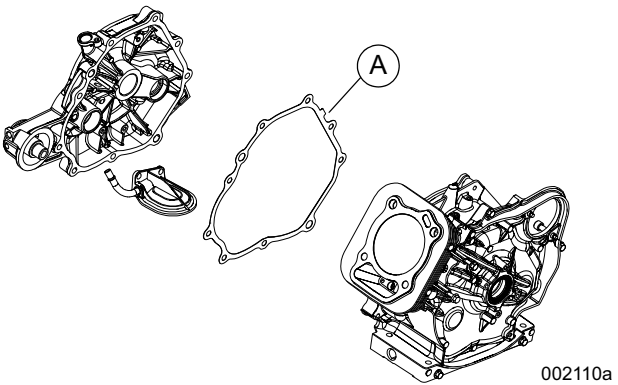


Figure 9-9. Assembly of Crankcase Cover and Oil Pump

- 5. Apply clean engine oil to crankshaft, camshaft, and balancer journals.
- 6. Apply clean engine oil to crankshaft, camshaft, and balancer bores in the crankcase cover.

NOTE: If engine is equipped with an optional oil filter, install filter adapter, gasket, and filter. If not equipped with an oil filter, install adapter gasket and oil block off cover.

- 7. Verify camshaft bore in crankcase cover and oil pump's inner rotor shaft have been coated with oil.
- 8. See [Figure 9-10](#). Apply a small amount of general purpose grease to the oil pump inner rotor (B). Grease will help prevent rotor from dropping out when the crankcase cover is inverted for installation.

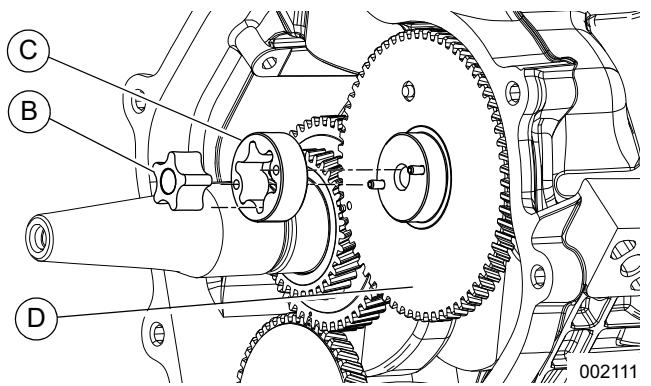


Figure 9-10. Location of Oil Pump Drive Pins

- 9. Install oil pump inner rotor over the rotor shaft and into the camshaft bore.
- 10. See [Figure 9-10](#). Install oil pump outer rotor (C) on two drive pins on the camshaft gear (D).

11. Carefully install crankcase cover. Align crankcase cover with dowel pins on the crankcase flange, as well as with crankshaft, camshaft, and balancer. Verify crankcase cover gasket is on. Rotate crankshaft to align oil pump inner and outer rotors. Crankcase cover should fall easily into place. DO NOT FORCE.
12. See [Figure 9-11](#). Install crankcase cover bolts and tighten. See [Specifications](#) for torque specifications.

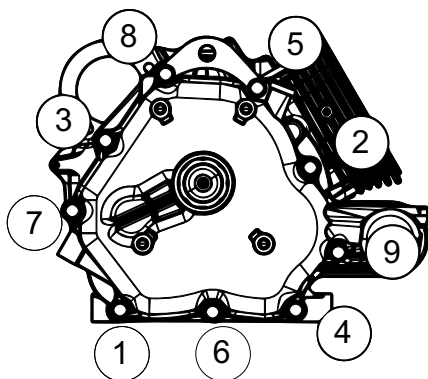


Figure 9-11. Crankcase Cover Bolt Tightening Sequence

13. Verify crankshaft turns freely after tightening crankcase cover bolts.

Oil Filter

See [Figure 9-12](#). The oil filter is a “spin-on” type. An oil filter adapter is bolted to the oil filter pad. When installing the oil filter, coat oil filter seal with engine oil, then install oil filter and tighten until seal contacts the filter adapter. After oil filter seal contacts filter adapter, tighten three-quarters to one full turn more.

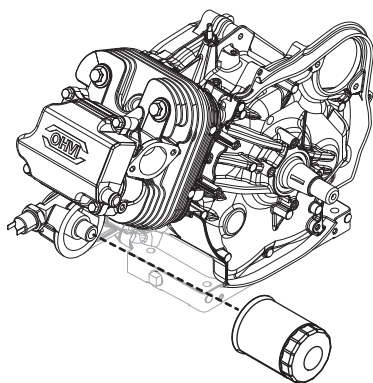
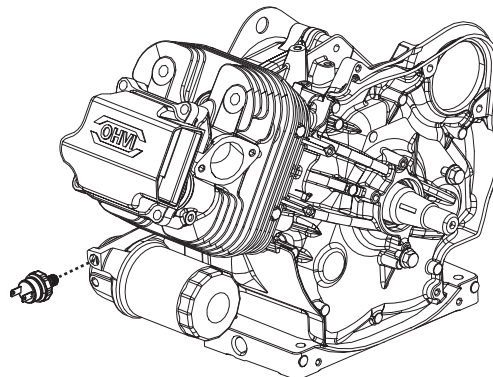


Figure 9-12. Oil Filter

Low Oil Pressure System

See [Figure 9-13](#). The engine is equipped with a low oil pressure switch.



001745

Figure 9-13. Low Oil Switch

If engine oil pressure drops below approximately 5 psi (34.5 kPa) for any reason, the switch contacts will close. With the switch contacts closed, the following will occur:

- A low oil pressure alert displays on the controller.
- The closure of switch contacts connects primary ignition circuit to ground. Ignition terminates, and the engine shuts down.

NOTE: Some differences may exist between low oil pressure systems. See electrical schematic/wiring diagram in the owner's manual for the specific application using this engine.

001744

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Section 10: Housing and Shrouds

General

Engine shrouds and housing need to be removed as part of the process of tearing down a complete engine for service and repairs.

Removal

Proceed as follows to remove housing and shrouds:

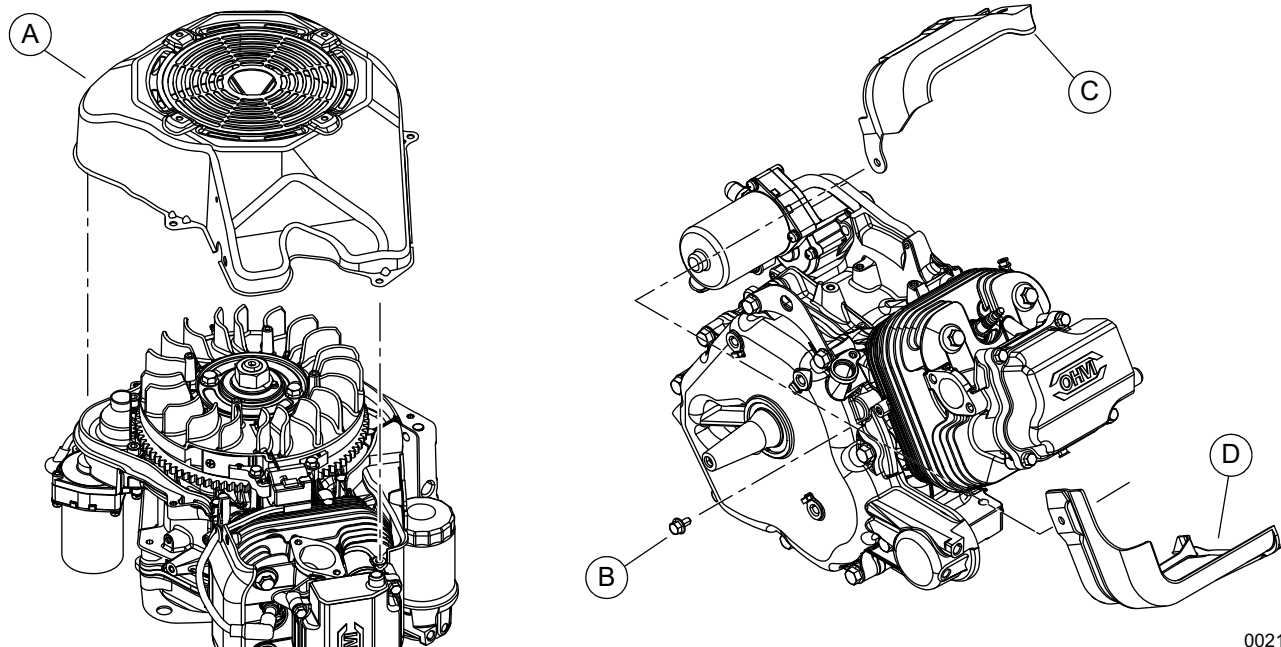
1. Remove air cleaner assembly and carburetor.
2. Disconnect governor rod and anti-lash spring from carburetor.
3. See [Figure 10-1](#). Remove the seven cap screws from blower housing assembly (A).
4. Carefully remove blower housing. Do not stretch the wires from the RUN/STOP switch and low oil pressure switch (LOPS). When the housing is far enough away from the engine, these wires may be unplugged.
5. Remove cap screw (B) from shrouds (C, D).

NOTE: The blower housing is notched at the point where the governor rod and anti-lash spring pass through it. This allows housing to be removed without disconnecting the governor rod from the governor arm.

NOTE: Make note of the locations of the screws that hold the covers in place.

Installation

Reverse previous steps to install the housing and shrouds. Verify wires for the RUN/STOP switch and LOPS are not in the way of the housing during reassembly.



002113

Figure 10-1. Exploded View of Engine Housing and Shrouds

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Section 11: Crankshaft, Camshaft, and Balancer

Crankcase Cover Removal

Remove all rust, burrs, and paint from the power take-off (PTO) end of the crankshaft before attempting to remove crankcase cover. This reduces the possibility of damaging the oil seal in the crankcase cover during cover removal. Proceed as follows to remove crankcase cover:

1. Drain oil from crankcase.
2. Remove engine cylinder head, push rods, and push rod guide plate (see [Cylinder Head and Valves](#)).
3. See [Figure 11-1](#). Remove all flange head bolts (A) retaining the crankcase cover, along with the lifting eye (B).

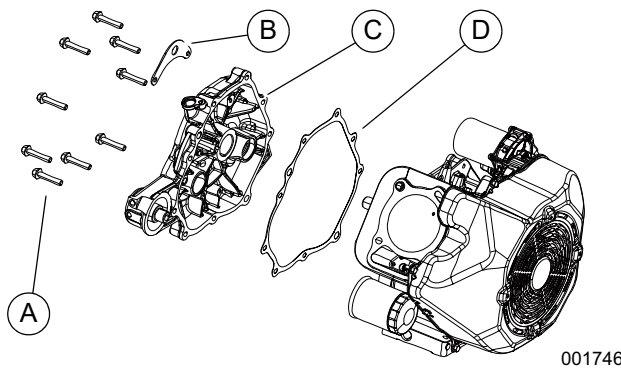
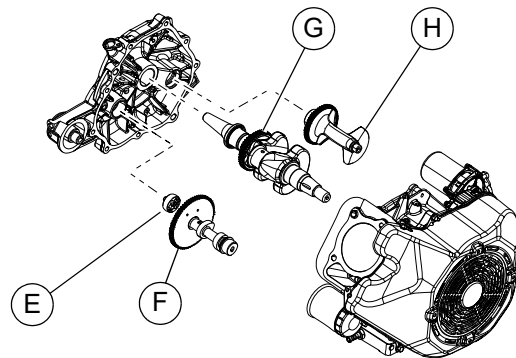


Figure 11-1. Crankcase Cover Removal

4. Remove crankcase cover (C) and gasket (D). If necessary, lightly tap with a soft hammer on alternate sides of cover. Be aware the oil pump rotors may drop out as the crankcase cover is removed.

Crankshaft Camshaft and Balancer Removal

See [Figure 11-2](#). With the crankcase cover removed, the oil pump (E), camshaft (F), crankshaft (G), and balancer (H) can be removed from their crankcase bores. Proceed as follows to remove these components:



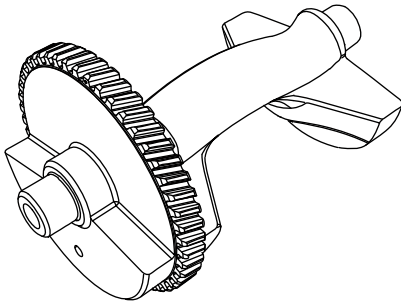
001747

Figure 11-2. Crankshaft, Camshaft, and Balancer Removal

1. Tip engine over onto flywheel end of crankshaft.
2. Reach in with two fingers and hold tappets up and clear of camshaft lobes. Remove camshaft, both tappets, camshaft spring washer, and inner and outer oil pump rotors.
3. Remove balancer.
4. Remove engine flywheel before removing crankshaft (see [Flywheel Removal](#)).
5. Rotate crankshaft to access the two connecting rod bolts.
6. Remove connecting rod bolts and connecting rod cap.
7. Push connecting rod and piston up and out of cylinder.

Balancer Inspection

See [Figure 11-3](#). Clean balancer in solvent. Then proceed as follows to inspect the balancer:



001749

Figure 11-3. Balancer

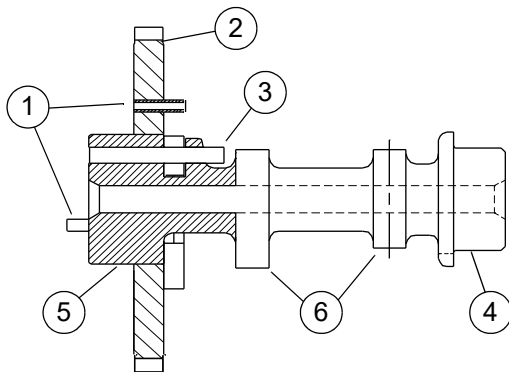
1. Inspect for physical damage. Replace balancer if damaged.
2. Measure outside diameter (O.D.) of balance shaft ends. Replace balancer if wear limits are exceeded.

NOTE: See [Specifications](#) for design sizes and wear limits.

3. Measure inside diameter (I.D.) of balancer shaft bearing bore in the crankcase cover and in the crankcase.
 - a. If balancer bearing bore in crankcase cover exceeds wear limits, replace crankcase cover. See [Specifications](#).
 - b. If balancer bearing bore in crankcase exceeds wear limits, replace crankcase assembly. See [Specifications](#).

Camshaft Inspection

Carefully inspect the entire camshaft for wear, nicks, or other damage. All areas indicated in [Figure 11-4](#) should be inspected.



002115

Figure 11-4. Points to Check on Camshaft

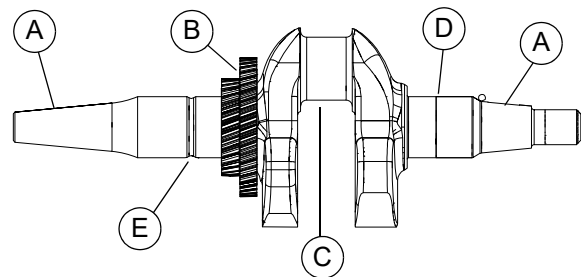
1	Spring pins
2	Camshaft gear
3	Compression release parts
4	Main bearing, flywheel end
5	Main bearing, PTO end
6	Cam lobes

Carefully measure both main bearing ends and inspect them for wear. Replace the faulty part if wear limits are exceeded. See [Specifications](#) for bearing wear limits.

Crankshaft Inspection

Crankshaft Body

Using a commercial solvent, clean the crankshaft thoroughly. Proceed as follows to inspect crankshaft after cleaning:



003805

Figure 11-5. Points to Check on Crankshaft

1. Inspect keyways (A) in crankshaft. Verify they are not worn or spread apart. Remove burrs from edges of keyways to prevent scratching the bearing.
2. Inspect crankshaft gears (camshaft and balancer) (B) for chipping or cracking. Replace crankshaft if timing gear is damaged.
3. Inspect crankpin (C) for nicks, scratches, or other damage. Small scratches and nicks can be polished out using fine emery cloth.

IMPORTANT NOTE: All emery cloth residue **MUST** be completely removed using a solvent such as kerosene.

4. Carefully measure diameters of crankpin, crankshaft journal at flywheel end (D), and crankshaft journal at PTO end (E). Replace crankshaft if worn beyond stated repair limits. See [Specifications](#) for crankshaft diameters.

NOTE: The crankpin must NOT be ground to any smaller diameter. Undersize connecting rods are NOT available.

Crankshaft Ball Bearing

See [Figure 11-6](#). The crankshaft is supported by a crankcase ball bearing (A) at one end. An oil seal (B) is seated in the bore on the other side of the crankcase.

NOTE: Assemble ball bearing on crankshaft before inserting crankshaft into crankcase.

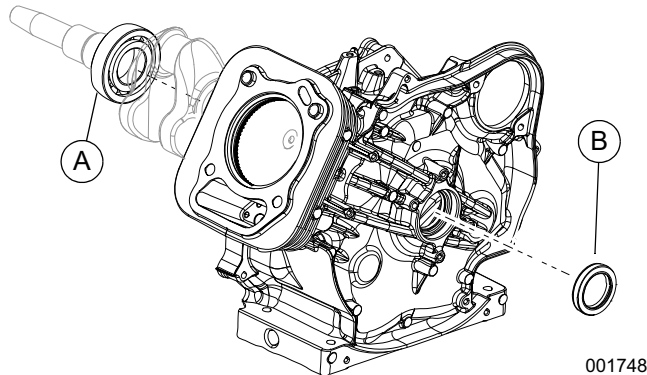


Figure 11-6. Crankshaft Bearings

The crankshaft bore in the crankcase cover also contains a pressure lubricated oil bearing.

Proceed as follows to inspect the crankshaft bore and ball bearing:

1. Inspect ball bearing for damage:
 - a. Measure inside diameter of ball bearing or bearing bore. Replace ball bearing if worn excessively.
 - b. Press out old bearing and press new bearing into place.

See [Specifications](#) for bearing and bore dimensions.

IMPORTANT NOTE: Alignment of oil holes in sleeve bearing and bearing bore in the crankcase is critical.

2. See [Figure 11-7](#). Inspect crankshaft bearing bore (A) in crankcase cover (B) for damage and wear.
 - a. Measure inside diameter of bearing bore.
 - b. If bore is worn excessively, replace crankcase cover.

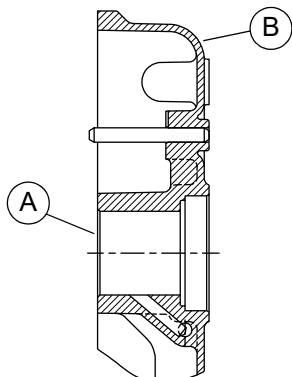


Figure 11-7. Bearing Bore in Crankcase Cover

Compression Release

A mechanical compression release is provided on the camshaft. A flyweight pivots (A) on a pin mounted in the camshaft gear. The pin itself has a cam action. When the engine is not running, the pin holds the tappet up just slightly which, in turn, holds the exhaust valve slightly open. This relieves compression for easier cranking.

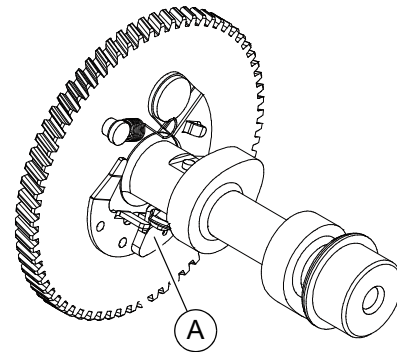


Figure 11-8. Compression Release Mechanism

When the camshaft spins, the flyweight moves, causing the pin to turn. The tappet is then free to move normally.

See [Figure 11-9](#). Measure the amount of compression relief lift at the tappet. See [Specifications](#) for measurements.

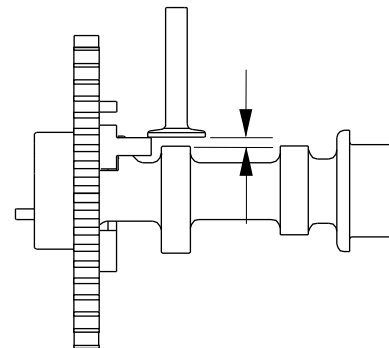
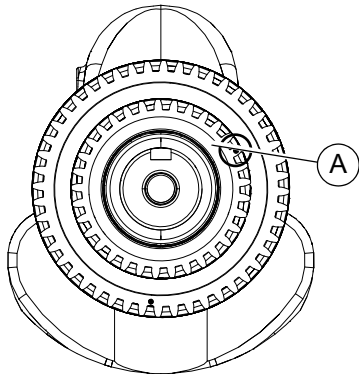


Figure 11-9. Compression Relief Lift

Crankshaft Installation

Seal protectors should be used to prevent damage to oil seals during installation. Proceed as follows to install the crankshaft:

1. Lubricate all bearing surfaces with engine oil.
2. Install valve tappets.
3. Support both ends of the crankshaft and carefully install into crankcase.
4. See [Figure 11-9](#). Notice timing mark (A) on crankshaft gear. Timing mark will be used to align crankshaft gears in the next step.



003909

Figure 11-10. Timing Mark on Crankshaft Gear

NOTE: A misaligned crankshaft gear will disrupt engine timing and permanently damage the crankshaft. On a fully assembled engine, the keyway on the end of the crankshaft should be in line with the center of the ignition coil when the piston is at TDC.

Camshaft Installation

Apply oil to camshaft bearing and to bearing bore in crankcase. Install camshaft in crankcase camshaft bore. Hold valve tappets out of the way during installation.

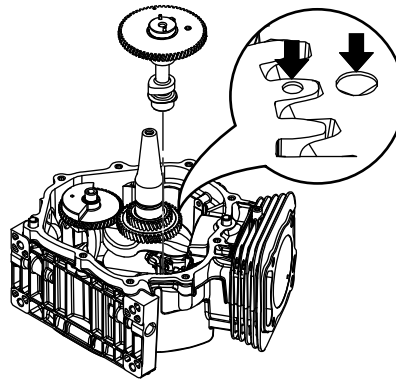
NOTE: See [Figure 11-11](#). All models are equipped with a camshaft spring washer (A). Use this washer during reassembly, and verify it is correctly seated between the camshaft and crankcase.



003844

Figure 11-11. Camshaft Spring Washer

See [Figure 11-12](#). Align timing mark on camshaft gear with timing mark on crankshaft gear. Piston must be at TDC.



002120

Figure 11-12. Alignment of Camshaft Timing Marks

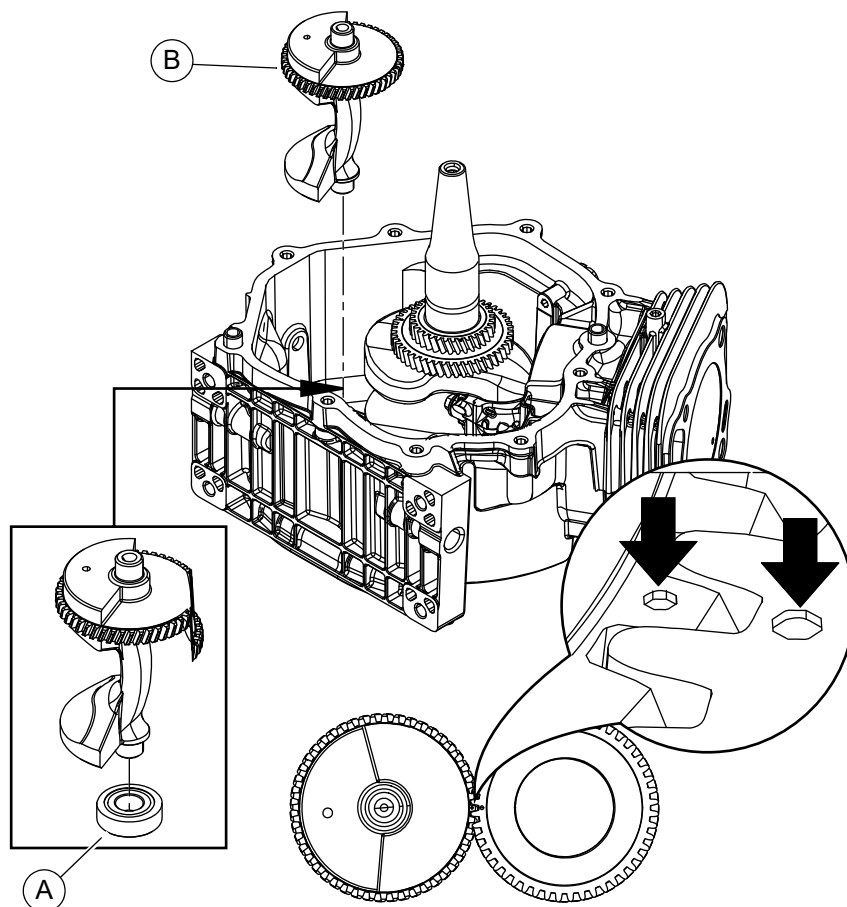
After installing crankshaft, the piston and connecting rod can be installed. (See [Piston, Rings, and Connecting Rod](#).)

Balancer Installation

See [Figure 11-13](#). Apply oil to balancer bearing surfaces and to bearing bore surfaces. Install balancer bearing (A) in the crankcase, and then install balancer (B) in the balancer bearing bore.

Balancer gear must mate with the crankshaft large gear and timing marks must be aligned as shown.

NOTE: For installation of oil pump, oil pickup, and crankcase cover, see [Lubrication System](#).



002121

Figure 11-13. Balancer Installation

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Section 12: Piston, Rings, and Connecting Rod

NOTE: There are no oversize pistons or rings available for these engines. For that reason, crankcase must be replaced if the cylinder is damaged or worn excessively.

Removing the Piston and Connecting Rod

See [Figure 12-1](#). Clean all carbon from the cylinder bore (to prevent ring breakage) before attempting to remove the piston (D) and connecting rod (A). Proceed as follows to remove piston and rod:

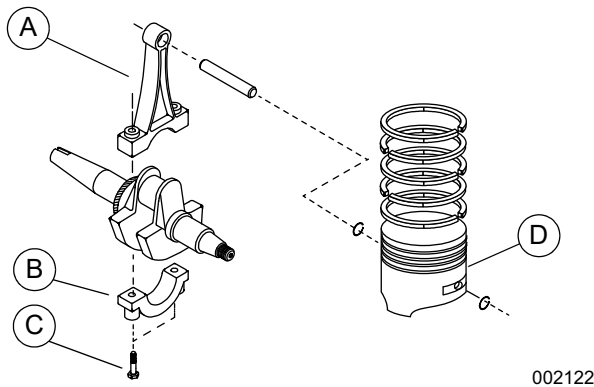


Figure 12-1. Pistons, Rings, and Connecting Rod

1. Remove cylinder head. See [Cylinder Head and Valves](#).
2. Remove crankcase cover. See [Lubrication System](#).
3. Remove connecting rod cap bolts (C) and connecting rod cap (B).
4. Push piston and rod out through top of cylinder.

Piston

Removing Piston From the Connecting Rod

See [Figure 12-2](#). An oil hole in the wrist pin (A) of the piston (B) helps distribute oil to aid in cooling. This oil hole is also provided to assist in removing the wrist pin snap ring (C).

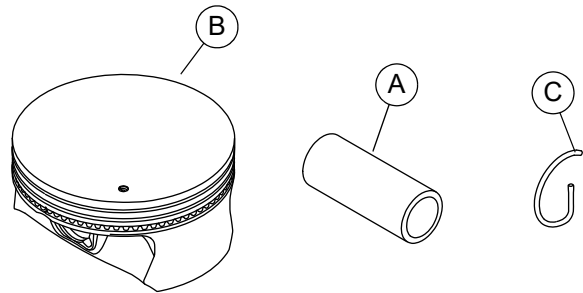


Figure 12-2. Piston, Wrist Pin, and Snap Ring

Proceed as follows to remove piston from rod:

1. Rotate snap ring until protruding wire is aligned with oil hole. Use needle nose pliers to turn snap ring and pull it toward you.
2. With one snap ring removed, slide wrist pin out of piston boss. Completely remove wrist pin and separate piston from the connecting rod.

Inspecting the Piston

The piston is slightly elliptical. Its major (larger) diameter (D) is perpendicular to the wrist pin boss; its minor (smaller) diameter (E) is in line with the wrist pin boss.

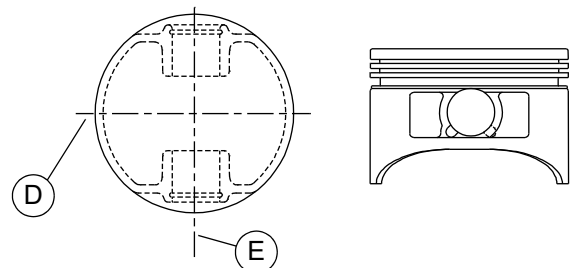


Figure 12-3. Piston Major and Minor Diameter

NOTE: An assembly mark dimple is provided on top of the piston. Assembly mark must face flywheel end of crankshaft (3:00 position) during reassembly.

Proceed as follows to inspect piston:

1. Measure piston's major diameter (D).
 - a. At a point 90° from the wrist pin hole, measure down from top of piston to a distance of 1.4–1.6 in (35.5–40.5 mm).
 - b. Measure at this point to check for wear. If wear is excessive, replace the piston. See [Specifications](#) for correct dimensions.
 2. Inspect wrist pin for looseness.
 - a. Inspect for wear in (a) the wrist pin, (b) wrist pin bore in piston, or (c) wrist pin bore in rod. Look for indications of a loose fit with the piston assembled to the rod.
 - b. A loose fit indicates a worn wrist pin, or a worn wrist pin bore in the piston or rod.
- NOTE:** Tolerances between the wrist pin and wrist pin bores are extremely close. Always apply engine oil to the pin and its bores prior to installation.
3. Inspect wrist pin and wrist pin bores for wear.
 - a. Measure outside diameter of wrist pin.
 - b. Measure inside diameter of wrist pin bore in piston.
 - c. Measure inside diameter of wrist pin bore in connecting rod.
 - d. Measure wrist pin length.
 4. Replace worn part if excessive wear is found. See [Specifications](#) for correct dimensions.

Piston Rings

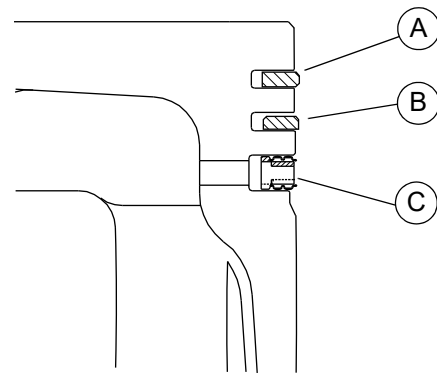
General Rules

The following general rules pertaining to piston rings must always be complied with:

- Always replace piston rings in sets.
- Use a ring expander to remove or install piston rings, or breakage could result. Do not spread the rings too far apart or they might break.
- Use a ring compressor when installing the piston in the cylinder.
- Deglaze cylinder walls with a commercially available deglazing tool when installing NEW rings.

Description

See [Figure 12-4](#). A piston ring set consists of a top compression ring (A), a second compression ring (B), and an oil control ring (C).



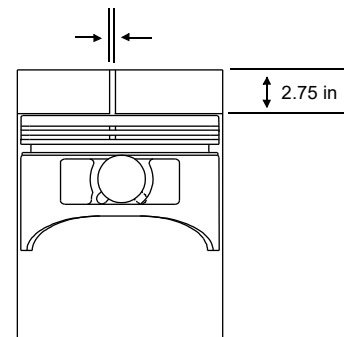
002125

Figure 12-4. Location of Rings in Piston Grooves

- The oil control ring is a three-piece assembly consisting of two oil rails and an oil spacer ring. Oil rails have a rounded face and may be installed with either side up.
- The second compression ring has an inside chamfer. This chamfer must face upward when installing the ring.
- The top compression ring has a barrel shaped face and can be installed with either side facing up.

Ring End Gap

See [Figure 12-5](#). Locate a point inside the cylinder 1.38 in (35 mm) down from the top of the cylinder (approximately halfway down). Place ring into cylinder and use piston to push ring down to the stated depth. Verify ring end gap with a feeler gauge.



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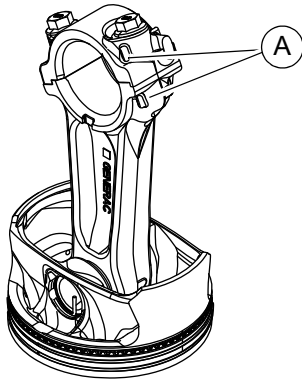
Figure 12-5. Measuring Ring End Gap

If the end gap of a new ring is excessive, the cylinder is worn excessively. See [Specifications](#) for correct dimensions.

NOTE: Oversize pistons and rings are not available. If the cylinder is worn or damaged, the crankcase must be replaced.

Connecting Rod

See [Figure 12-6](#). The connecting rod is manufactured of die-cast aluminum. Match marks (A) on the rod and on rod cap must be aligned when assembling the rod to the crankshaft.



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Figure 12-6. Match Marks on Rod and Cap

IMPORTANT NOTE: Do not use washers when assembling the rod cap. Added thickness could cause rod cap to become loose or interfere with closely-fitted parts, resulting in catastrophic engine failure.

Assembly and Installation

See [Assembly](#) section for complete instructions.

Cylinder Service

Inspection

Proceed as follows to inspect the cylinder:

1. Inspect cylinder for dirty, broken, or cracked cooling fins.
2. Inspect cylinder wall for wear, scoring, or other damage.
3. Inspect all bearing surfaces for wear, scoring, or other damage.

Reboring the Cylinder

IMPORTANT NOTE: Do not attempt to rebore the cylinder. Oversize parts are not available.

Oversize pistons and rings are not available. It is therefore impossible to rebore the cylinder for purposes of retrofitting oversize parts. Replace engine if cylinder is worn.

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Section 13: Troubleshooting

Problem Solving

Introduction

Problems affecting engine operation may be classified as one of, or a combination of two or more of the following:

- Engine will not start
- Engine starts with difficulty
- Engine lacks power
- Engine vibrates
- Engine overheats
- Engine burns oil

In many cases, the cause of a problem will be readily apparent. If this is not the case, inspect the fuel system, ignition system, and engine compression. These inspections can be performed quickly and are the best methods of finding the cause of a failure. They may also provide early detection of future failures.

Checking Fuel System

NOTE: If using in home standby product, use generator diagnostic manual for troubleshooting fuel system.

Verify fuel tank has been correctly filled with the correct fuel. Verify fuel shutoff valve is open, if equipped, and fuel is flowing through the fuel line and to the carburetor. Adjust carburetor as needed. Verify choke closes completely. Remove and inspect spark plug if engine still does not start.

If plug is WET, look for the following:

- Overchoking
- Dirty air cleaner
- Excessively rich fuel-air mixture
- Water in fuel
- Fuel leaking past the carburetor float needle

If plug is DRY, look for the following:

- Carburetor mounting gasket leaks
- Dirt or gum in carburetor or fuel line
- Carburetor float needle stuck closed
- Inoperative fuel pump (if equipped)
- Clogged fuel filter (if equipped)

Checking Ignition System

See [Ignition](#).

Checking Compression

Proceed as follows to check engine compression:

1. Remove spark plug.
2. Insert an automotive type compression gauge into spark plug hole.
3. Crank engine until there is no further increase in pressure. The highest reading obtained is the engine compression pressure.

If compression is poor, inspect for one or more of the following causes:

- Loose cylinder head bolts
- Failed cylinder head gasket
- Burned valves or valve seats
- Insufficient valve clearance
- Warped cylinder head
- Warped valve stem
- Worn or broken piston ring(s)
- Worn or damaged cylinder bore
- Broken connecting rod
- Worn valve seats or valves
- Worn valve guides

Problem	Possible Causes	Remedy
Engine will not start or starts with difficulty	Empty fuel tank.	Fill fuel tank.
	Low oil level.	Replenish oil as necessary.
	Dirty air cleaner.	Clean or replace air cleaner element.
	Obstructed fuel line.	Clean or replace fuel line.
	Obstructed fuel tank cap vent hole.	Open fuel tank cap vent hole.
	Fouled spark plug.	Replace spark plug.
	Incorrect spark plug.	Replace with correct spark plug.
	Loose or faulty ignition wiring.	Inspect wiring; repair or replace.
	No ignition spark.	Inspect ignition system; replace faulty part(s).
	Incorrect valve clearance.	Reset valve clearance.
	Poor compression.	Inspect for worn or scored cylinder, possible burnt valve, or burnt valve seat.
Engine will not start, or starts and runs rough	Fuel shutoff valve is OFF.	Set fuel shutoff valve to ON.
	Dirty air filter.	Clean or replace air filter.
	Insufficient fuel.	Fill fuel tank.
	Stale fuel.	Drain fuel tank and fill with fresh fuel.
	Water in fuel.	
	Spark plug wire not connected to spark plug.	Connect wire to spark plug.
	Faulty spark plug.	Replace spark plug.
	Over-choking.	Set choke to no choke.
	Insufficient oil level.	Fill crankcase to correct level.
	Excessive rich fuel mixture.	Contact an IASD.
	Intake valve stuck open or closed.	
	Engine compression lost.	
	Dirty fuel filter.	Replace fuel filter.
	Poor speed signal.	Check for AC voltage between pin 1 and pin 2 on carb connector.
Engine knocks	Carbon in combustion chamber.	Clean carbon from head and piston.
	Loose flywheel.	Inspect flywheel key and keyway; replace parts if necessary. Tighten flywheel nut.
	Worn cylinder.	Replace crankcase.
	Loose or worn connecting rod.	Replace connecting rod. Verify crankpin diameter to see if crankshaft is faulty.
	Incorrect valve clearance.	Reset to correct clearance.
	Overloaded engine.	Reduce excessive load.

Problem	Possible Causes	Remedy
Engine runs well at no load, but bogs when load is applied	Short circuit in a connected load.	Disconnect shorted electrical load.
	Generator is overloaded.	See Know Generator Limits in the owner's manual.
	Engine speed is too low.	Contact an IASD.
	Shorted generator circuit.	
	Dirty fuel filter.	Replace fuel filter.
	Clogged spark arrestor.	Clean spark arrestor screen.
	Speed is slow, throttle not opening.	Check for 12V between pin 1 and pin 3 on carb connector.
Engine shuts down during operation	Insufficient fuel.	Replenish fuel.
	Insufficient oil level.	Fill crankcase to correct level.
	Fault in engine.	Contact an IASD.
	Throttle closed.	Check carb for throttle operation.
Engine misses under load	Incorrect spark plug.	Install correct spark plug.
	Spark plug is fouled.	Replace spark plug.
	Spark plug porcelain is cracked.	
	Incorrect spark plug gap.	Regap spark plug.
	Incorrect valve clearance.	Reset valve clearance.
	Weak valve springs.	Replace weak valve spring(s).
Engine lacks power	Governor set incorrectly.	Adjust governor.
	Incorrect spark plug.	Install correct spark plug.
	Incorrect valve clearance.	Reset valve clearance.
	Worn piston rings.	Replace piston rings.
	Low oil level.	Replenish oil to correct level.
	Air cleaner is obstructed.	Clean or replace air cleaner.
	Valves or valve seats worn or burned.	Grind valves and valve seats.
	Excessive load.	Reduce load. See Know Generator Limits in the owner's manual.
	Dirty air filter.	Clean or replace air filter.
	Engine maintenance required.	Contact an IASD.
	Dirty fuel filter.	Replace fuel filter.
	Clogged spark arrestor.	Clean spark arrestor screen.
	Speed is slow, throttle not opening.	Check for 12V between pin 1 and pin 3 on carb connector.

Problem	Possible Causes	Remedy
Engine is running, but AC output is not available	Circuit breaker is OPEN.	Reset circuit breaker.
	Poor connection or faulty cord set.	Inspect cord set and repair.
	Connected device is faulty.	Connect another device that is in good condition.
	Fault in generator.	Contact an IASD.
	GFCI breaker module is OPEN.	Correct ground fault and press reset button on GFCI breaker module.
Engine overheats	Air flow is obstructed.	Remove obstructions.
	Cooling fins are clogged.	Clean cooling fine.
	Carbon buildup in combustion chamber.	Remove cylinder head and clean carbon.
	Engine is overloaded.	Reduce excessive loading.
	Lack of lubrication.	Replenish oil to correct level.
Engine vibrates excessively	Engine mounting bolts loose.	Tighten mounting bolts.
	Bent crankshaft.	Replace crankshaft.
	Problem in equipment connected to engine.	Inspect equipment driven by engine.
Engine surges or runs unevenly	Vent hole in fuel tank cap plugged.	Open vent hole in fuel tank cap.
	Governor parts sticking or binding.	Clean governor parts and replace as necessary.
	Carburetor throttle linkage, throttle shaft binding.	Clean, lubricate, adjust, or repair sticking parts.
	Intermittent ignition spark.	Test ignition as per Ignition section; repair or replace faulty parts as necessary.
	Incorrect carburetor adjustment.	Adjust carburetor.
	Dirty carburetor.	Clean carburetor.
	Choke opened too soon.	Set choke to halfway position until engine runs smoothly.
	Carburetor is running too rich or too lean.	Contact an IASD.
	Intermittent speed signal.	Check for AC voltage between pin 1 and pin 2 on carb connector.
Engine overspeeds	Throttle is open.	Check for 12V to carb.
	No speed signal.	Check for AC voltage between pin 1 and pin 2 on carb connector.

Problem	Possible Causes	Remedy
Engine uses excessive amounts of oil	Engine speed too fast.	Adjust engine speed to specifications.
	Oil level too high.	Drain excess oil from crankcase.
	Oil filler cap loose, or cap gasket damaged.	Replace cap gasket and tighten.
	Breather damaged or dirty.	Clean or replace breather as necessary.
	Drain holes in breather clogged.	Unclog drain holes in breather.
	Gaskets or seals leaking.	Replace gaskets or seals.
	Valve guides worn excessively.	Replace cylinder head.
	Cylinder walls worn.	Replace crankcase.
	Piston rings and/or grooves worn.	Install new rings and/or piston.
	Oil passages obstructed.	Clean out oil passages.
Crankshaft oil seal leaks	Crankcase breather dirty or faulty.	Clean or replace breather.
	Oil seal hardened or worn.	Replace oil seal.
	Crankshaft seal contact surface is worn.	Check crankshaft size and replace if worn.
	Crankshaft sleeve bearing worn.	Replace sleeve bearing.
	Seal not seated squarely in bore.	Replace with new seal.
	New seal damaged during installation.	
	Bent crankshaft.	Inspect crankshaft for straightness; replace if necessary.
Breather is passing oil	Engine speed too fast.	Adjust engine speed to specifications.
	Oil level too high.	Drain excess oil from crankcase.
	Oil filler cap loose, or cap gasket damaged.	Replace cap gasket and tighten cap.
	Breather damaged or dirty.	Clean or replace breather as necessary.
	Drain holes in breather clogged.	Unclog drain holes in breather.
	Piston ring end gaps are aligned.	Rotate ring end gaps, stagger 90°.
	Rings seated incorrectly.	Inspect for worn or "out of round" cylinder. Install new rings. Break in new rings with engine under varying loads.

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Section 14: Assembly

Introduction

The following engine assembly procedure is provided for reference. Instructions apply to engines for both home standby (HSB) and portable generators, except where noted.

NOTE: To avoid damaging the engine or causing leaks, use correct techniques when removing and installing gaskets. See [Gasket Removal](#).

Installing Crankcase Bearing and Seal

Proceed as follows to install crankcase bearing and seal:

1. See [Figure 14-1](#). Inspect crankcase (A) to verify governor hole (B) is not plugged. Verify oil level switch wire hole plug (C) is in place. Hole plug edges should be flush with crankcase surface.

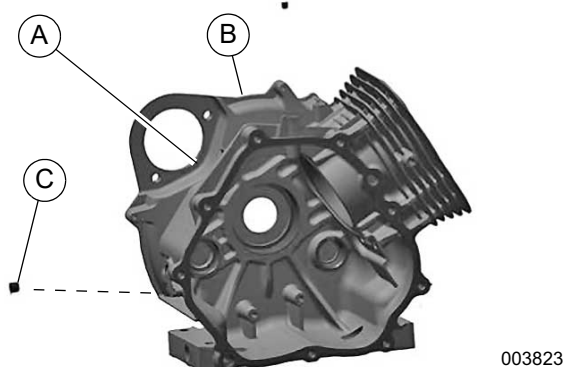


Figure 14-1. Base Crankcase Assembly

2. See [Figure 14-2](#). Verify crankcase side balancer bearing bore (D) is clean and free of burrs or debris. Press crankcase side balancer bearing (E) into bore until fully seated.

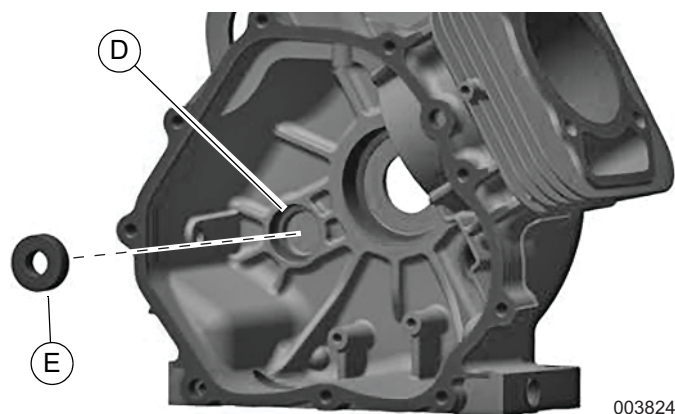


Figure 14-2. Install Crankcase Bearing Bore

3. See [Figure 14-3](#). Verify crankcase side main seal bore (F) is clean and free of burrs or debris. Orient crankcase side main seal (G) with cavity toward crankcase. Press seal into crankcase until fully seated.

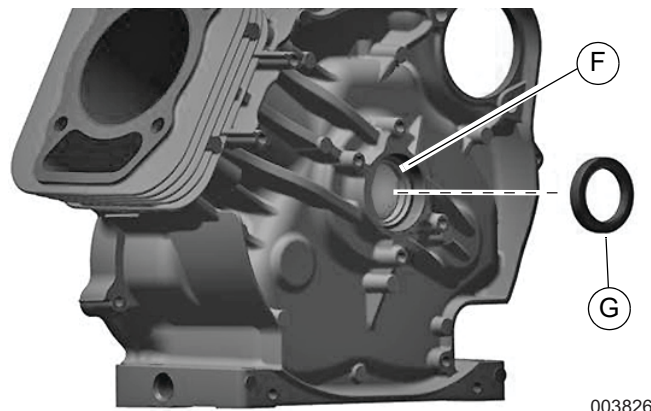


Figure 14-3. Install Crankcase Side Main Seal

Assembling Internal Governor (Portable)

Proceed as follows to assemble the internal governor for portables:

1. See [Figure 14-4](#). Place governor washer (A) on governor rod (B). Slide governor rod through hole in crankcase.

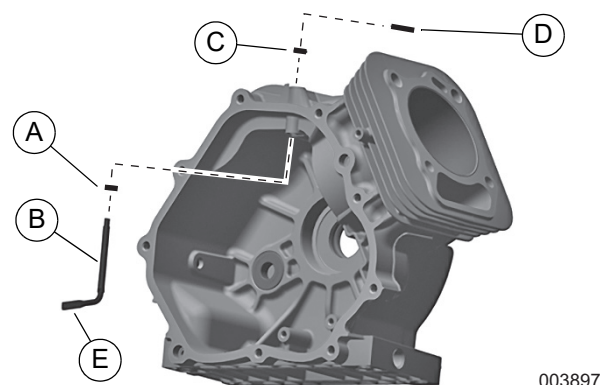


Figure 14-4. Install Governor Rod

2. Place nylon washer (C) on governor rod. Secure governor rod with governor clip (D) by passing clip through small hole in the rod.
3. Verify governor rod paddle (E) points towards rear of crankcase, away from cylinder.

Assembling Crankshaft

Proceed as follows to assemble the crankshaft:

1. See [Figure 14-5](#). Verify camshaft gear (A) and balancer drive gear (B) are on the crankshaft (C) with teeth aligned as shown.

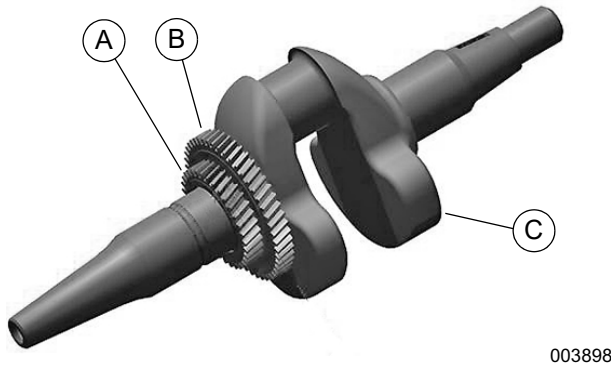


Figure 14-5. Crankshaft With Gears

2. See [Figure 14-6](#). Verify timing marks on balancer gear and cam drive gear are correctly aligned on crankshaft.

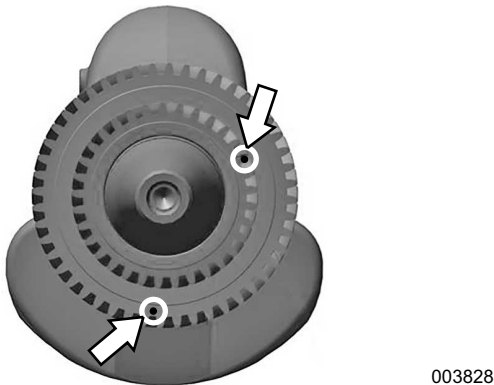


Figure 14-6. Timing Mark Alignment

3. See [Figure 14-7](#). Verify crankshaft and main bearing mating surface (D) are clean and free of burrs and debris. Press main bearing (E) onto crankshaft until fully seated against mating surface.

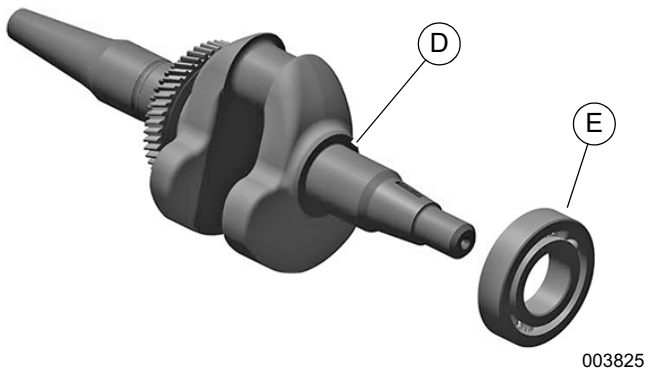


Figure 14-7. Main Bearing Installation

4. Apply Generac SAE30 engine oil to balls in main bearing.

Installing Crankshaft Assembly

Proceed as follows to install the crankshaft assembly:

1. See [Figure 14-8](#). Apply assembly lube to entire surface of crankshaft connecting rod journal (A).

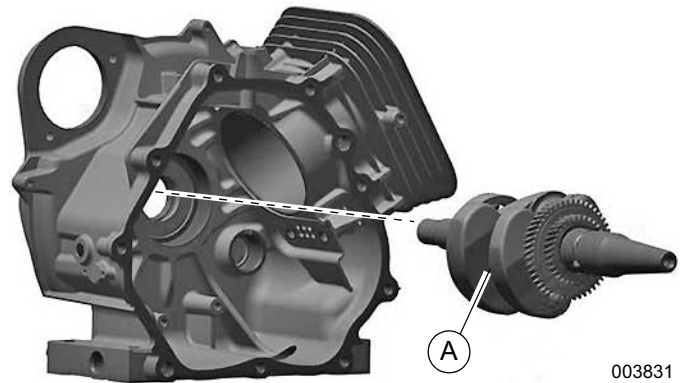


Figure 14-8. Install Crankshaft

2. Install crankshaft assembly in crankcase as shown, with cam and balancer drive gears on open side of case assembly.
3. See [Figure 14-9](#). Main bearing (B) must be fully seated in crankcase bore.

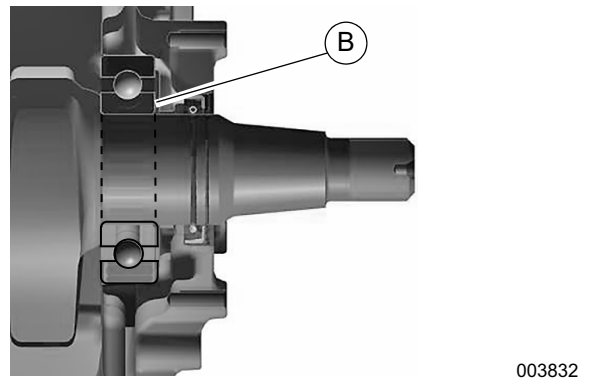
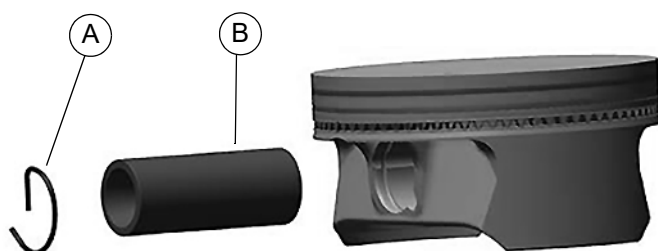


Figure 14-9. Main Bearing Seat

Assembling Piston

Proceed as follows to assemble the piston:

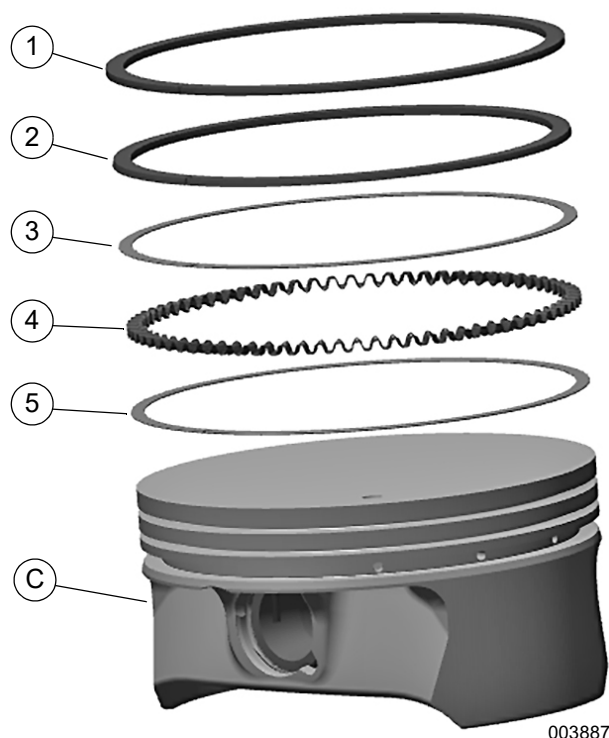
1. See [Figure 14-8](#). Remove one piston pin retainer clip (A) and piston pin (B). Set aside for future use.



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Figure 14-10. Piston Retainer Clip and Pin

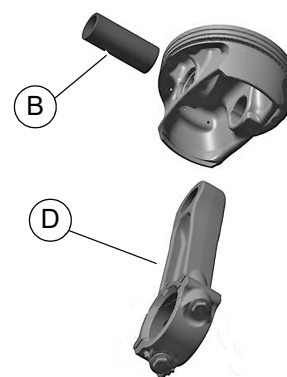
2. See [Figure 14-11](#). Install piston rings (1, 2, 3, 4, 5) on piston (C) with dots toward the top of the piston. Piston ring gaps will be aligned as part of procedure [Installing Connecting Rod](#).



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Figure 14-11. Install Piston Rings

3. See [Figure 14-12](#). Apply assembly lube to piston pin (B). Place piston on connecting rod (D), with part number on connecting rod facing gear cover. Install piston pin.

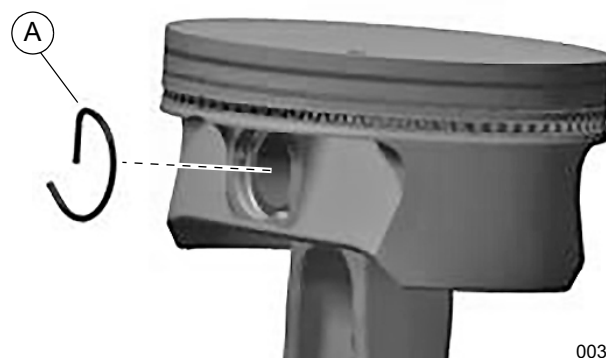


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Figure 14-12. Install Piston

NOTE: Arrow on piston crown and dot on piston point to flywheel side of the engine.

4. See [Figure 14-13](#). Secure piston with retainer clip (A). Verify clip is fully seated in groove in piston pin bore.



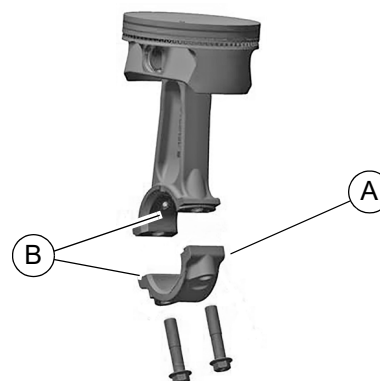
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Figure 14-13. Install Retaining Clip

Installing Connecting Rod

Proceed as follows to install connecting rod:

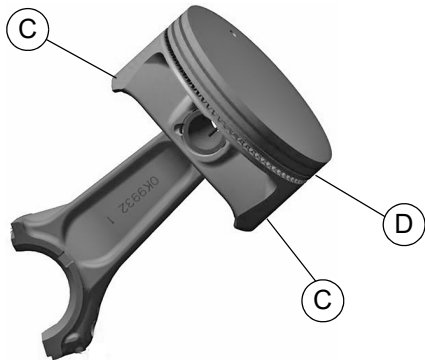
1. See [Figure 14-14](#). Clamp connecting rod assembly in the soft jaws of a vise with the large end facing up. Remove connecting rod cap (A).



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Figure 14-14. Connecting Rod and Cap

2. Apply assembly lube to bearing surfaces (B) of connecting rod and connecting rod cap.
3. See [Figure 14-15](#). Apply Generac 30W engine oil to outer surfaces of piston (C), rings (D), and cylinder bore in crankcase.



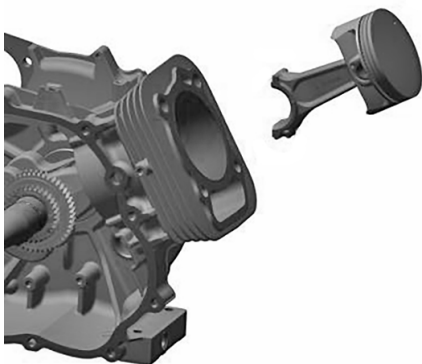
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Figure 14-15. Oiling Piston

4. See [Figure 14-11](#). Align piston ring gaps as follows:
 - Top ring (1): gap at 12 o'clock position
 - Second ring (2): gap at 6 o'clock position
 - Top oil ring (3): gap at 10 o'clock position
 - Oil expander ring (4): gap at 2 o'clock position
 - Bottom oil ring (5): gap at 4 o'clock position
5. See [Figure 14-16](#). Insert connecting rod assembly into cylinder as shown. Fit large end of connecting rod over crankshaft connecting rod journal.

NOTE: Avoid scratching journal surface.

NOTE: Arrow on piston crown and dot on piston point to flywheel side of the engine.

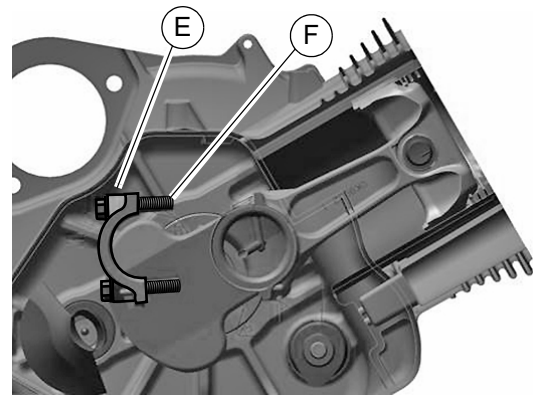


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Figure 14-16. Insert Piston

6. See [Figure 14-17](#). Install connecting rod cap (E) and connecting rod bolts (F). Tighten connecting rod bolts to 18 ft-lbs (24.4 Nm).

NOTE: Avoid scratching journal surface.



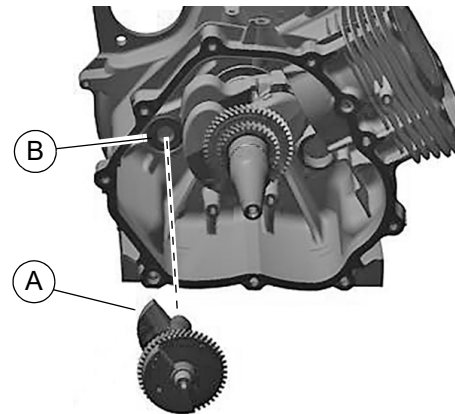
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Figure 14-17. Install Connecting Rod Cap

Installing Balancer Shaft

Proceed as follows to install balancer shaft:

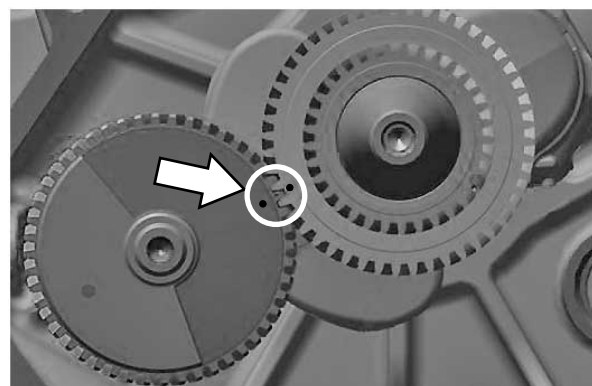
1. See [Figure 14-18](#). Insert balancer shaft (A) into balancer bearing (B) in the crankcase assembly as shown.



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Figure 14-18. Install Balancer Shaft

2. See [Figure 14-19](#). Align balancer and crankshaft balancer drive gear timing marks.



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Figure 14-19. Align Timing Marks

Installing Tappets

See [Figure 14-20](#). Apply assembly lube to the cylindrical portion of both tappets (A). Install them in the crankcase assembly.

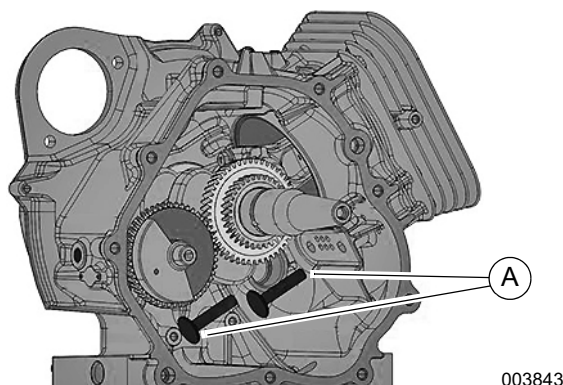


Figure 14-20. Install Tappets

Installing Camshaft Assembly

Proceed as follows to install the camshaft assembly:

1. See [Figure 14-21](#). Apply assembly lube to both sides of camshaft bevel washer (A). Install bevel washer on camshaft assembly (B).

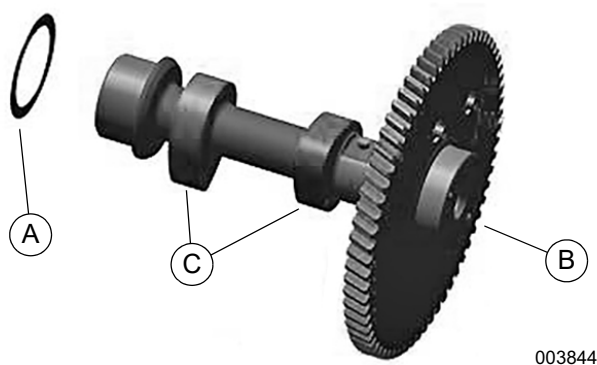


Figure 14-21. Install Bevel Washer

2. Verify both tappets are fully recessed in the crankcase. This protects camshaft lobes (C) from damage during cam installation.
3. See [Figure 14-22](#). Apply assembly lube to camshaft lobes and case side bearing journal (D). Install camshaft assembly in crankcase.

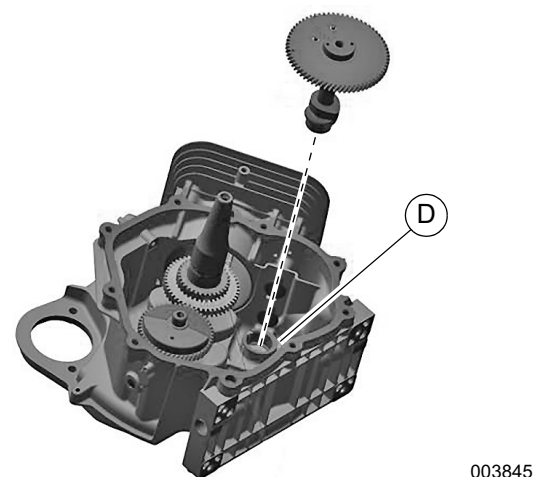


Figure 14-22. Install Camshaft Assembly

4. See [Figure 14-23](#). Align timing marks on camshaft driven gear and crankshaft cam drive gear.

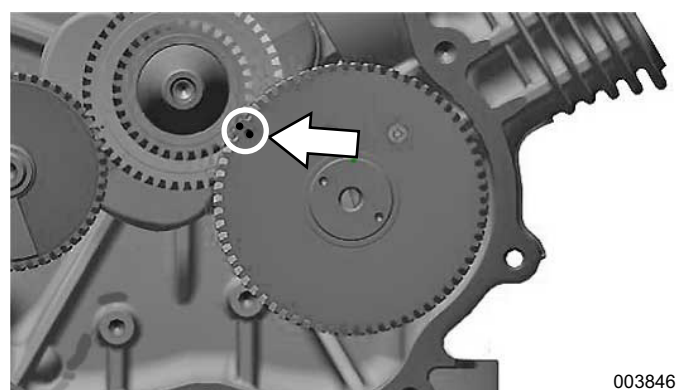


Figure 14-23. Align Timing Marks

Installing Oil Pump Gerotor

Proceed as follows to install the oil pump gerotor:

1. See [Figure 14-24](#). Apply a thin film of assembly lube to oil pump gerotor set (A). Install oil pump gerotor pins (B) and oil pump gerotor set.

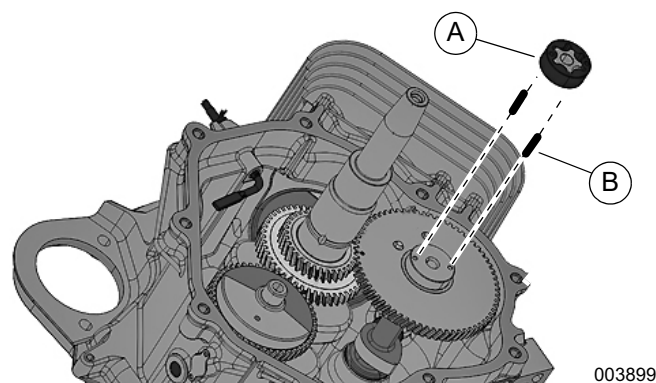


Figure 14-24. Install Oil Pump Gerotor

2. See [Figure 14-25](#). Install cover dowels (C) and oil pump inner oil pump gerotor dowel (D) in case and cam assemblies, respectively. Coat all surfaces of the oil pump inner oil pump gerotor dowel with a thin film of assembly lube.

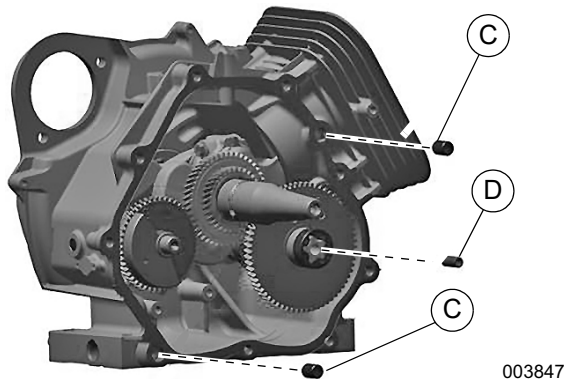


Figure 14-25. Install Dowels

Installing Gear Cover

Proceed as follows to install gear cover:

1. See [Figure 14-26](#). Install gear cover gasket (A) on crankcase mating surface. Verify gasket is aligned with dowels and seated correctly.

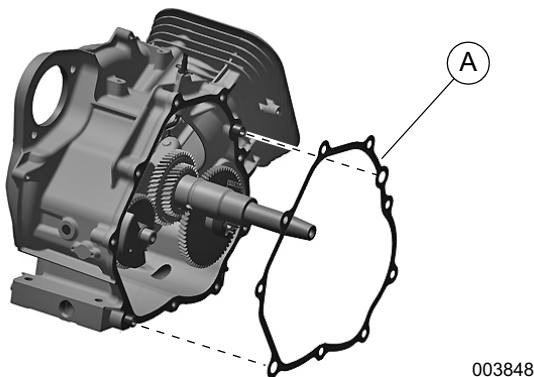


Figure 14-26. Install Gear Cover Gasket

2. See [Figure 14-27](#). Verify gear cover side balancer bearing bore (B) and gerotor bore (C) are clean and free of burrs and debris.

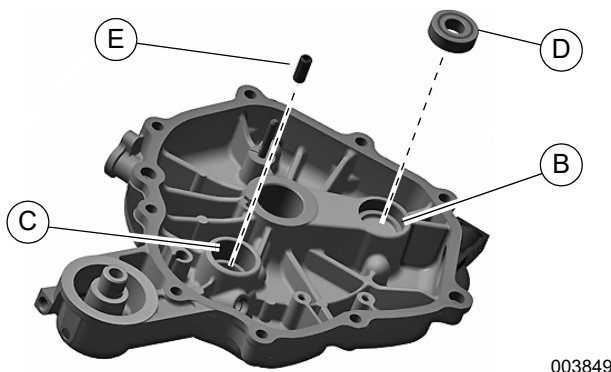


Figure 14-27. Balancer Bearing and Gerotor Dowel

3. Press balancer bearing (D) into gear cover until flush with cast surface on the cover. Press gerotor dowel (E) into hole in gerotor bore until fully seated.
4. Apply assembly lube to gerotor bore.
5. See [Figure 14-28](#). Verify PTO seal bore (F) is clean and free of burrs and debris. Press crankshaft PTO seal (G) into gear cover until it contacts bottom of PTO seal bore.

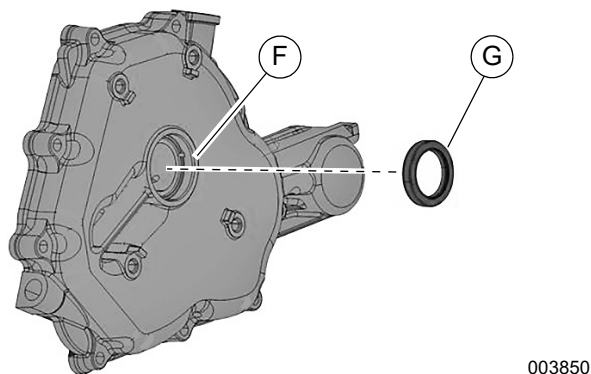


Figure 14-28. Install PTO Seal

Installing Oil Components

Proceed as follows to install oil components:

1. See [Figure 14-29](#). Verify pressure relief valve bore (A) is clean and free of burrs and debris. Assemble oil pressure relief valve by first inserting ball (C), then spring (D), then cover (E), then fastener (F). Tighten fastener to 44 in-lbs (5.0 Nm).

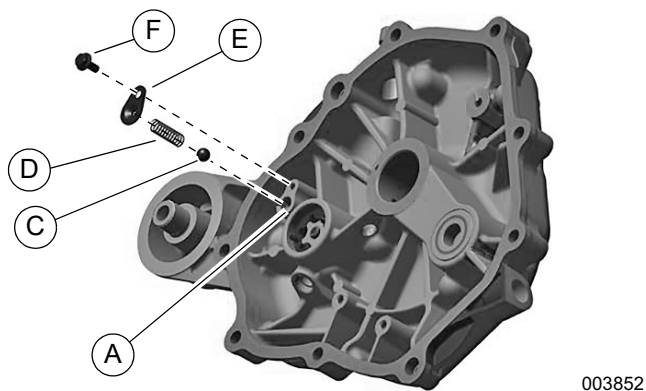


Figure 14-29. Assemble Pressure Relief Valve

- See [Figure 14-30](#). Verify oil pick-up tube (G) is clean and free of burrs and debris. Install two O-rings (H).

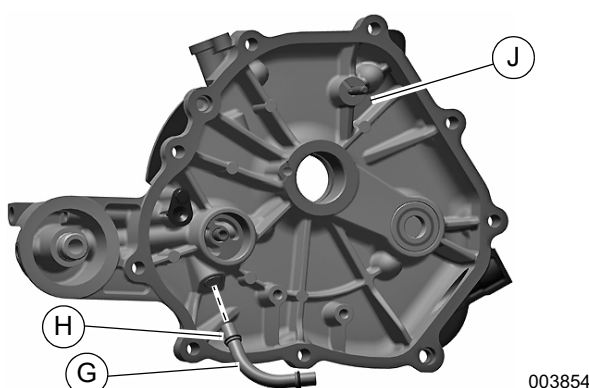


Figure 14-30. Install Oil Pick-Up Tube

- Apply oil to O-rings. Install oil pick-up tube assembly by inserting the paint-marked end into gear cover.
- (Portable only)** Insert governor shaft (J) into governor shaft hole.
- See [Figure 14-30](#). Install oil pump pick-up (K) on gear cover assembly. Insert loose end of oil pump pick-up tube into the pick-up as shown.

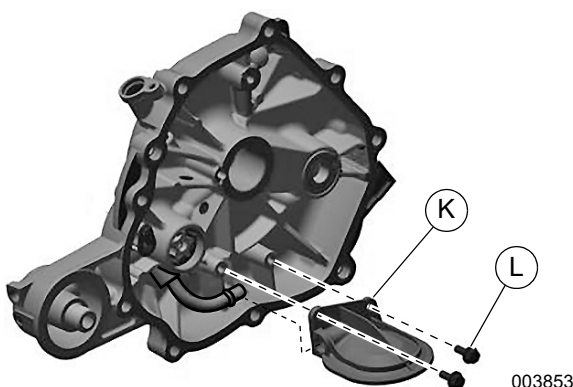


Figure 14-31. Install Oil Pump Pick-Up

- Tighten fasteners (L) to 6.6 ft-lbs (9.0 Nm).

Installing Internal Governor (Portable)

Proceed as follows to install internal governor in portable units:

- See [Figure 14-32](#). Build governor gear assembly by installing the two flyweights (A) in governor gear (B). Retain each flyweight by inserting a flyweight pin (C) through holes in gear and base of flyweights.

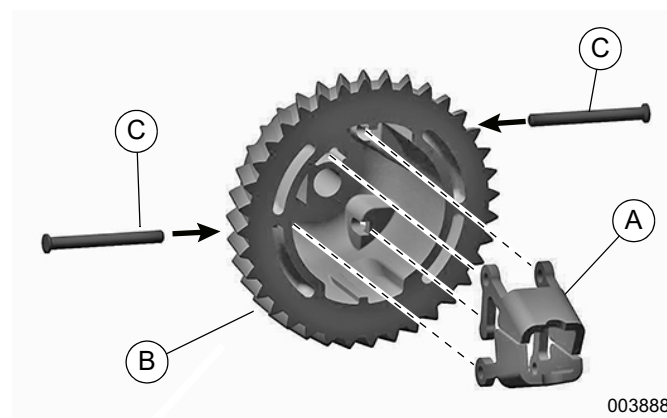


Figure 14-32. Governor Gear Assembly

- See [Figure 14-33](#). Slide governor washer (D), governor gear assembly, retaining c-clip (E), and governor spool (F) over governor shaft (G).

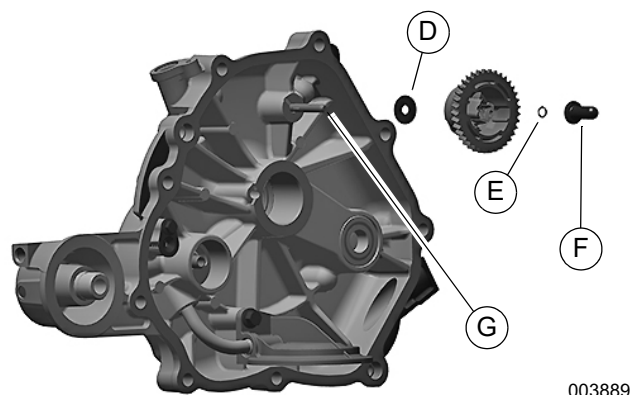


Figure 14-33. Install Internal Governor

- See [Figure 14-34](#). Verify the following:
 - governor washer (D) is seated on gear cover (H)
 - governor assembly (J) is seated on washer
 - retaining c-clip (E) is seated against governor gear (K)
 - spool (L) is seated on flyweight levers

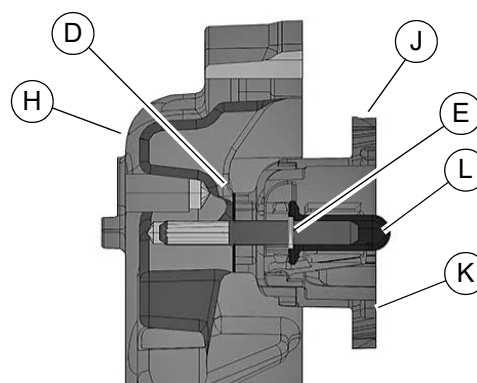


Figure 14-34. Installed Governor

Installing Gear Cover Assembly

Proceed as follows to install gear cover assembly:

1. See [Figure 14-35](#). With crankcase assembly placed on flywheel side (not shown), install gear cover assembly using nine fasteners (A) and lift hook (B; located at top two holes).

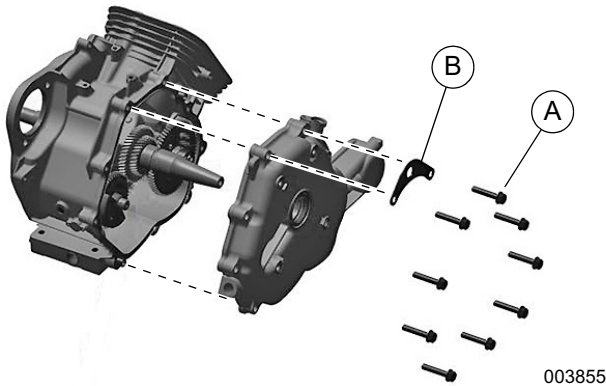


Figure 14-35. Install Gear Cover Assembly

2. Verify crankshaft, dowels, oil pump, and balancer line up with corresponding locations in gear cover. If cover will not seat, rotate crankshaft until cover drops into place.
3. See [Figure 14-36](#). Tighten the nine fasteners to 20 ft-lbs (27 Nm) in specified torque sequence.

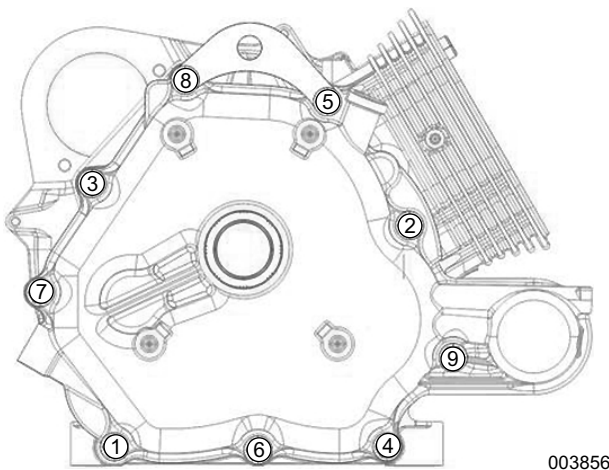


Figure 14-36. Fastener Torque Sequence

Installing Valves and Valve Springs

Proceed as follows to install valves and valve springs:

1. See [Figure 14-37](#). Install intake and exhaust valve seals (A) on head assembly (B).

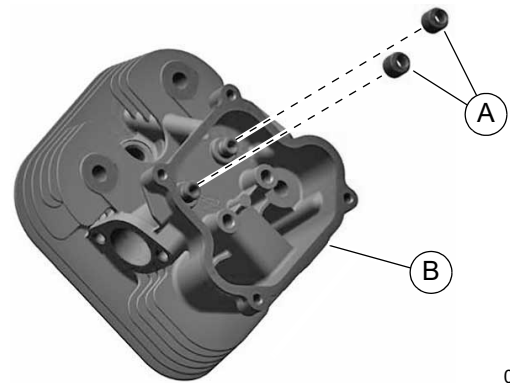


Figure 14-37. Install Valve Seals

2. See [Figure 14-38](#). Apply assembly lube to intake valve stem (C). Insert stem into valve guide (D) in head assembly at the combustion chamber side.

NOTE: Avoid damaging valve seal while inserting valve stem.

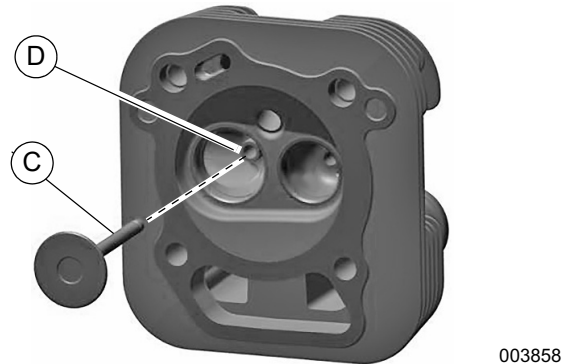


Figure 14-38. Insert Intake Valve

3. See [Figure 14-39](#). Place valve spring lower washer (E) over guide (F), then install valve spring (G) and upper collar (H). Verify washer is fully seated on cast surface of cylinder head, with valve spring and upper collar fully seated and aligned atop washer.

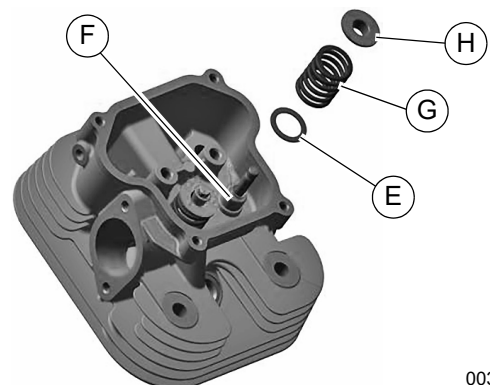


Figure 14-39. Install Valve Spring Assembly

4. Using the same procedure described in steps 2 and 3, insert exhaust valve and install exhaust valve spring assembly.
5. See [Figure 14-40](#). Using a valve spring compressor, compress each valve spring assembly enough to allow installation of valve spring keepers (J; two per valve). Verify each keeper inner rib is seated in the valve stem groove and locked in place. Remove valve spring compressor.

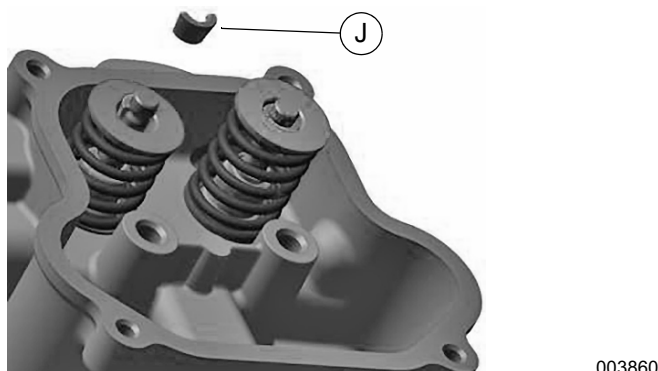


Figure 14-40. Install Valve Spring Keepers

Installing Head Assembly

Proceed as follows to install head assembly:

1. See [Figure 14-41](#). Install cover dowels (A) in crankcase. Verify cylinder head surface (B) is clean and free of burrs and debris.

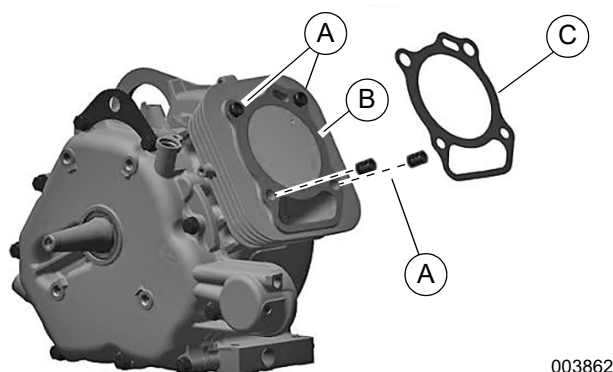


Figure 14-41. Install Dowels and Head Gasket

2. Install head gasket (C) on cylinder head surface. Verify gasket is seated correctly and aligned with dowels and holes in crankcase.
3. See [Figure 14-42](#). Install cylinder head assembly (D) on crankcase using four cylinder head bolts (E) and cylinder head bolt washers (F).

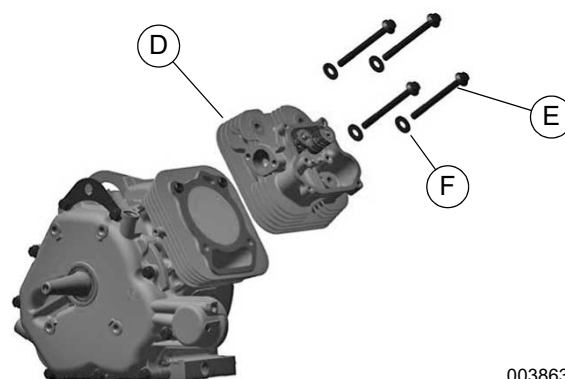


Figure 14-42. Install Cylinder Head

4. See [Figure 14-43](#). Tighten cylinder head bolts to 11.1 ft-lbs (15 Nm) in specified torque sequence.

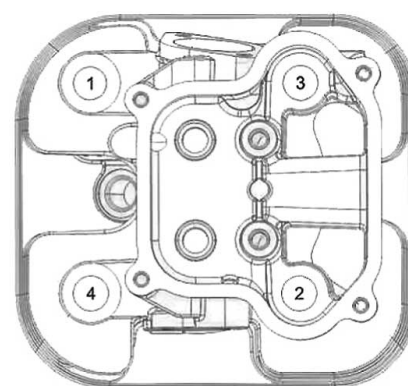


Figure 14-43. Cylinder Head Bolt Tightening Sequence

5. Repeat tightening sequence and tighten bolts to 33.2 ft-lbs (45 Nm).

Installing Rocker Arms

Proceed as follows to install rocker arms:

1. See [Figure 14-44](#). Apply assembly lube to each end of both pushrods (A). Insert pushrods into tappet recess on intake and exhaust side. Verify pushrods are completely seated in tappet recesses.

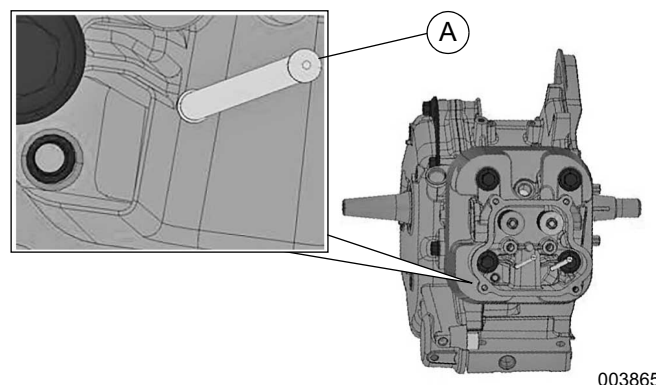


Figure 14-44. Install Pushrods

2. See [Figure 14-45](#). Install pushrod guide plate (B) rocker stud (C), rocker adjusting ball (D), rocker jam nut (E), and rocker arm (F) for both the intake and exhaust valves. Verify ball ends of pushrods are seated in recesses at bottom of rocker arms.

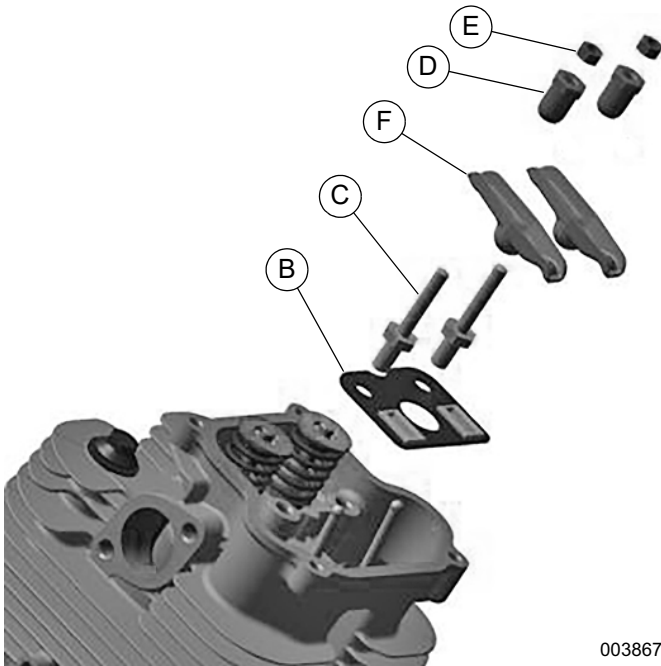


Figure 14-45. Assemble Rocker Arms

3. Tighten rocker studs to 9.2 ft-lbs (12.5 Nm).
4. See [Figure 14-46](#). Insert a 0.003 in (0.076 mm) shim between rocker arm and intake valve stem.

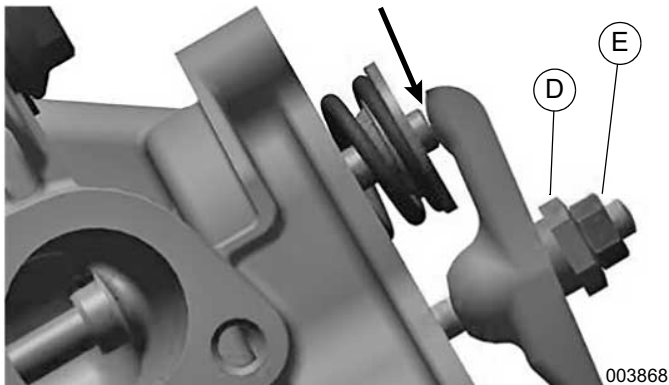


Figure 14-46. Adjust Intake Valve Clearance

5. Tighten rocker adjusting ball (D) until resistance is felt when removing shim.
6. Tighten jam nut (E) to 10 ft-lbs (13.5 Nm) while holding rocker adjusting ball to prevent it from rotating.
7. Repeat steps 4–6 to adjust exhaust valve clearance.

Installing Rocker Cover

See [Figure 14-47](#). Install rocker cover gasket (A), rocker cover assembly (B), and four fasteners (C). Tighten fasteners to 6.6 ft-lbs (9.0 Nm).

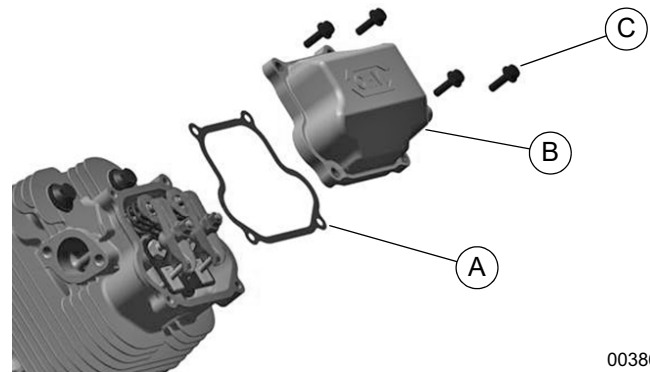


Figure 14-47. Install Rocker Cover

Installing External Components

Installing Thermal Switch (HSB)

See [Figure 14-48](#). Apply thread sealant to (2) fasteners (A). Install thermal switch (B) at crankcase rear. Tighten fasteners to 8.9 in-lbs (1.0 Nm).

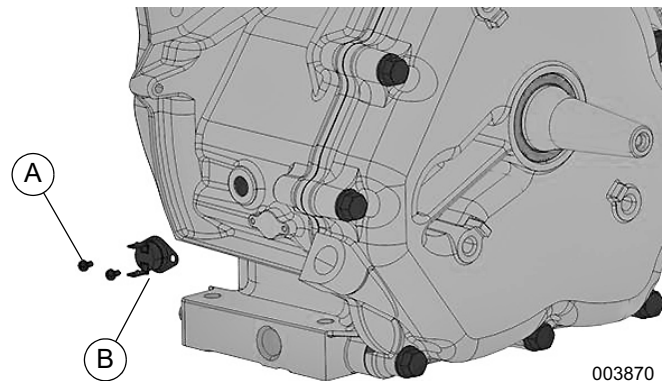


Figure 14-48. Install Thermal Switch (HSB)

Installing Oil Pressure Switch

See [Figure 14-48](#). Apply thread sealant to oil pressure switch (A) threads. Install switch at front of oil filter mount (B). Tighten oil pressure switch to 7.0 ft-lbs (9.5 Nm).

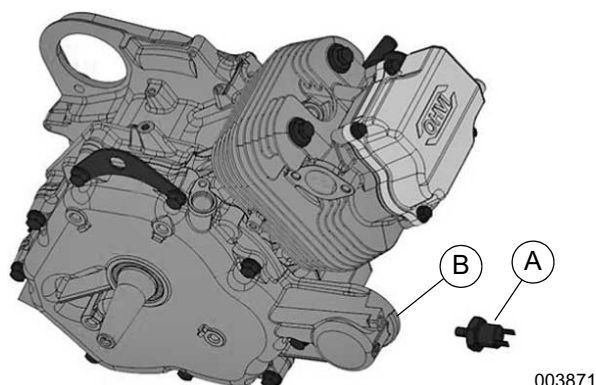


Figure 14-49. Install Oil Pressure Switch

Installing Oil Dipstick (Portable)

See [Figure 14-50](#). Place o-ring (A) on dipstick (B). Apply a small amount of 5W-30 oil to o-ring. Insert dipstick into crankcase and hand tighten.

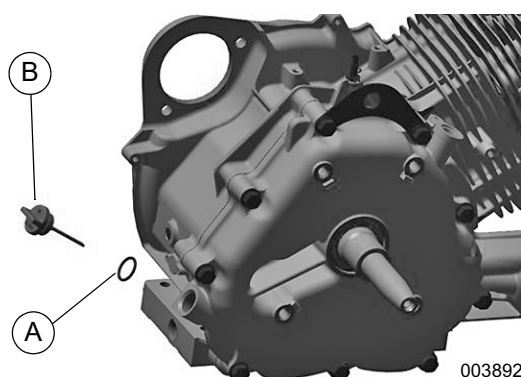


Figure 14-50. Install Oil Dipstick (Portable)

Installing Oil Dipstick (HSB)

See [Figure 14-51](#). Place o-ring (A) on dipstick tube (B). Apply a small amount of 5W-30 oil to o-ring.

Insert assembly into crankcase and fasten with screw (C). Tighten screw to 6.6 ft-lbs (9.0 Nm). Insert dipstick (D).

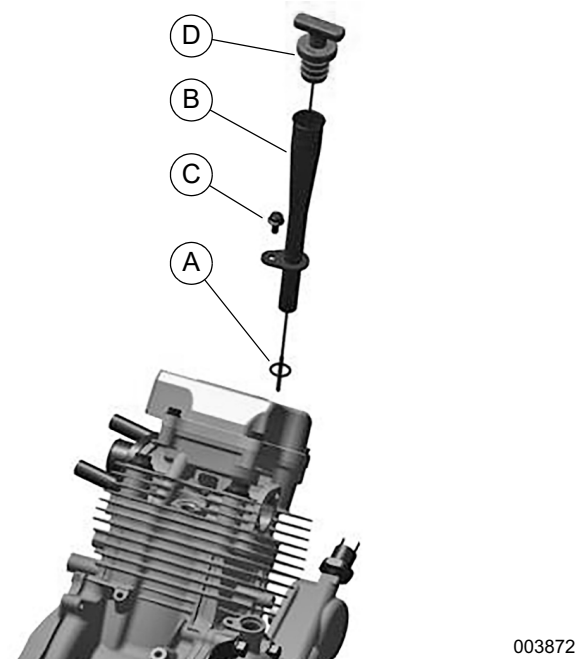


Figure 14-51. Install Oil Dipstick (HSB)

Installing Oil Drain Hardware

See [Figure 14-52](#). The engine is equipped with two oil drain ports (A: front and rear).

- **Portable:** Both ports are plugged.
- **HSB:** Rear port is plugged. Front plug is equipped with an oil drain hose.

Portable

Apply thread sealant to each port plug (B). Insert plugs in ports and tighten to 10 ft-lbs (13.5 Nm).

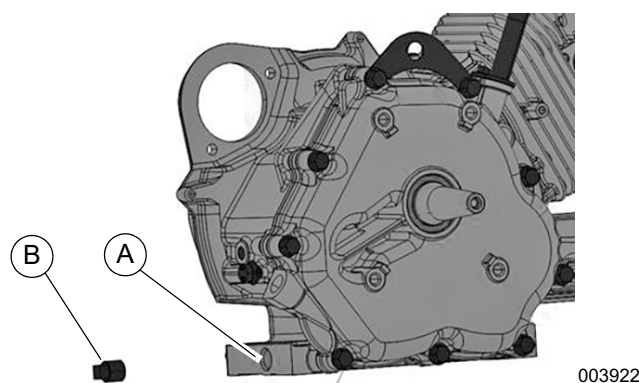


Figure 14-52. Install Oil Drain Plug

HSB

Proceed as follows to install oil drain hardware for HSB units:

1. Insert plug in rear drain port as described above.
2. See [Figure 14-53](#). Apply thread sealant to hose fitting (C) and install in front drain port (A). Tighten fitting to 10.0 ft-lbs (13.5 Nm).

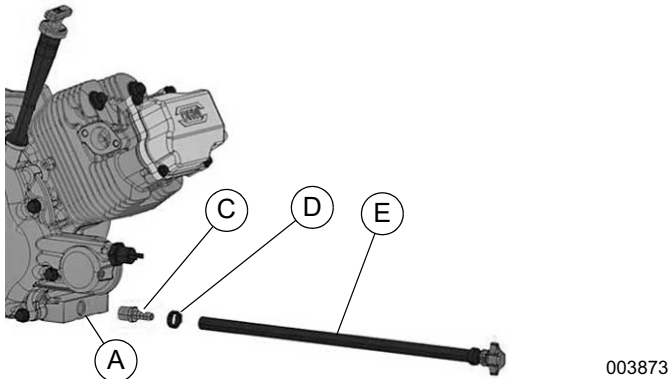


Figure 14-53. Install Oil Drain Hose (HSB)

3. Place clamp (D) over open end of the drain hose assembly (E). Push open end of hose onto fitting until fully seated. Secure with clamp.

Installing Oil Filter

See [Figure 14-54](#). Apply a thin coating of 30W oil to o-ring on oil filter (A). Install oil filter on filter mount at the front of the engine. Tighten oil filter to 5.2 ft-lbs (7.1 Nm).

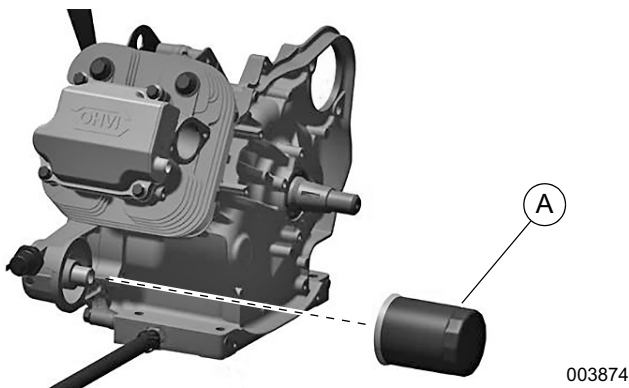


Figure 14-54. Install Oil Filter

Assembling Flywheel and Fan

Portable

Proceed as follows to assemble flywheel and fan for portable units:

1. See [Figure 14-57](#). Attach fan (A) to flywheel (B) with fan plate (C) and two fasteners (D). Tighten fasteners to 18.4 ft-lbs (25 Nm).

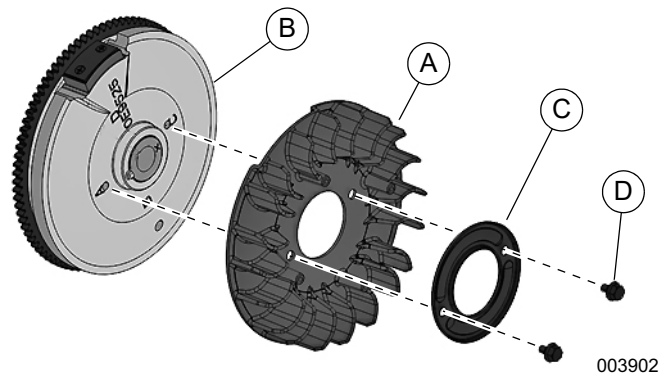


Figure 14-55. Attach Fan to Flywheel (Portable)

2. See [Figure 14-56](#). Verify crankshaft end is clean and free of burrs and debris. Install woodruff key (E) in crankshaft slot.

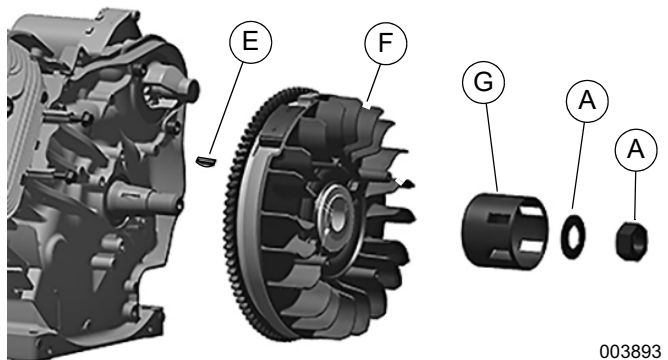


Figure 14-56. Attach Flywheel To Crankshaft (Portable)

3. Slide flywheel/fan assembly (F) and recoil cup (G) over crankshaft end. Secure with washer (H) and flywheel nut (J). Tighten flywheel nut to 90 ft-lbs (122.0 Nm).

HSB

Proceed as follows to assemble flywheel and fan for HSB:

1. See [Figure 14-57](#). Verify crankshaft end (A) is clean and free of burrs and debris. Install woodruff key (B) in crankshaft slot.

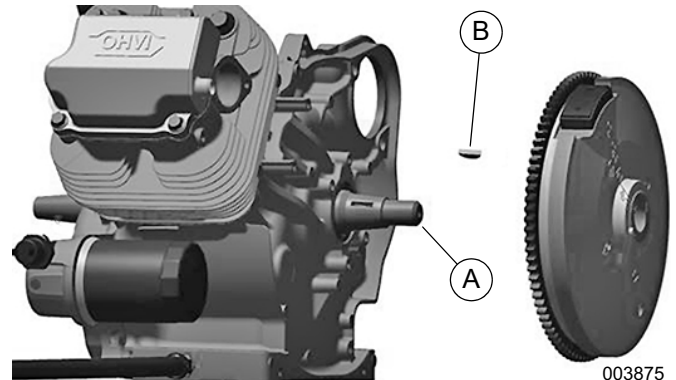


Figure 14-57. Install Woodruff Key (HSB)

- See **Figure 14-58**. Attach flywheel (C) to crankshaft with washer (D) and nut (E). Orient washer with the crown on the nut side.

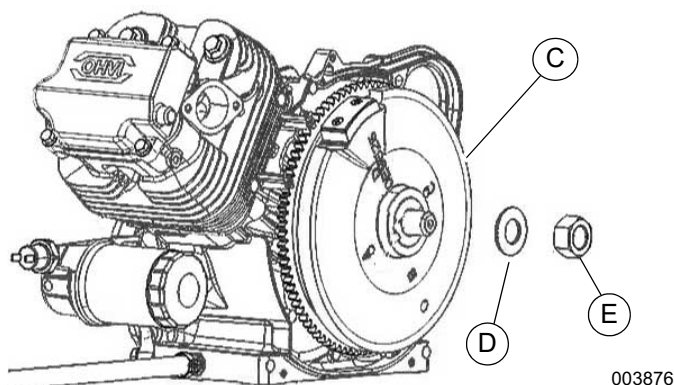


Figure 14-58. Attach Flywheel (HSB)

- Tighten flywheel nut to 90.0 ft-lbs (122.0 Nm).
- See **Figure 14-59**. Attach fan (F) to flywheel using fan plate (G) and two fasteners (H). Tighten fasteners to 18.1 ft-lbs (24.5 Nm).

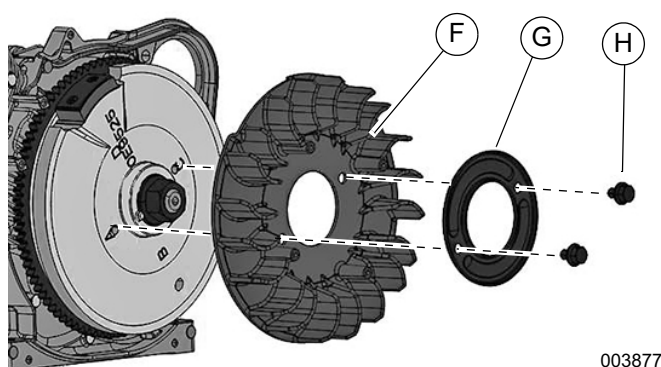


Figure 14-59. Attach Fan (HSB)

Assembling Ignition

Proceed as follows to assemble ignition:

- See **Figure 14-60**. Install spark plug (A) in spark plug hole. Tighten spark plug to 18.4 ft-lbs (25.0 Nm).

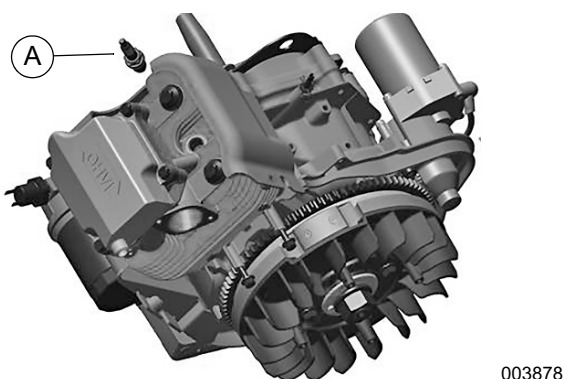


Figure 14-60. Install Spark Plug

- See **Figure 14-61**. Install ignition coil and spark plug wire assembly (B) on engine at 10 o'clock position above the flywheel when facing flywheel. Use two fasteners (C).

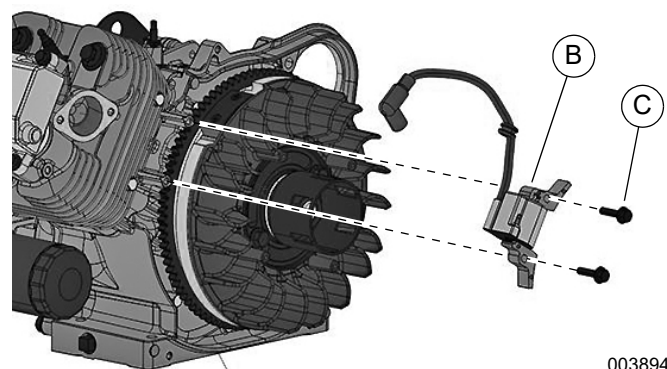


Figure 14-61. Install Ignition Coil

- Set air gap to 0.010 in (2.54 mm) and tighten fasteners to 6.6 ft-lbs (9.0 Nm).

NOTE: Attach free coil wire end to top of spark plug. An audible snap indicates wire is attached correctly.

Assembling Shroud and Wrapper

Proceed as follows to assemble shroud and wrapper:

- See **Figure 14-62**. Install starter assembly (A) using two fasteners (B). Tighten fasteners to 20.3 ft-lbs (27.5 Nm).

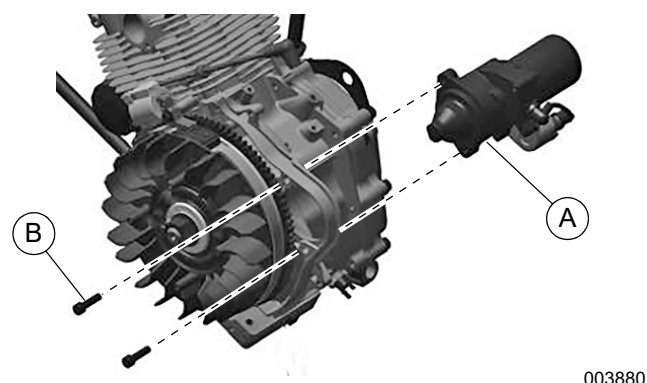


Figure 14-62. Install Starter

- See **Figure 14-63**. Position tab (C) on top wrapper beneath bump-out (D) on bottom wrapper as shown. Fasten wrappers to crankcase with one screw (E). Tighten screw to 6.6 ft-lbs (9.0 Nm).

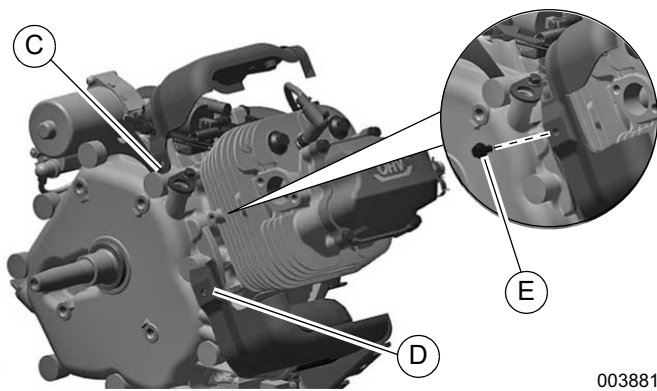


Figure 14-63. Install Wrappers

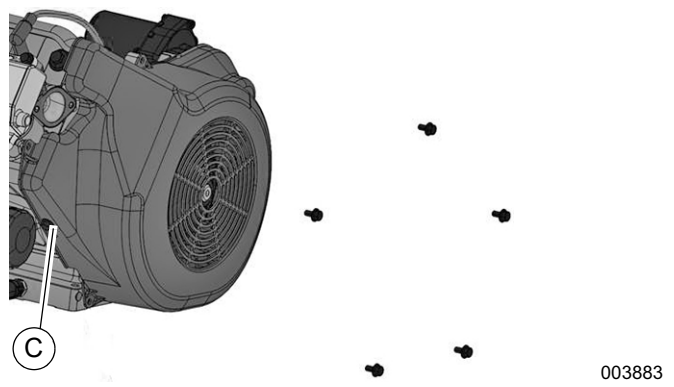


Figure 14-66. Install Blower Housing Screws

Installing Blower Housing

Proceed as follows to install blower housing:

1. See [Figure 14-64](#) (portable) or [Figure 14-65](#) (HSB). Verify coil wire grommet is firmly located in top wrapper slot (A) before installing blower housing (B). Install blower housing on engine, securing it to wrappers and engine with two screws (C).

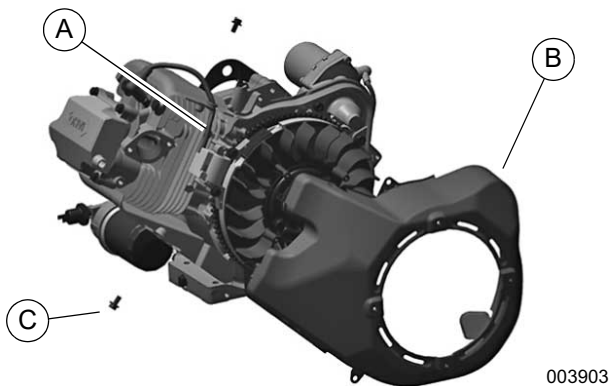


Figure 14-64. Install Blower Housing (Portable)

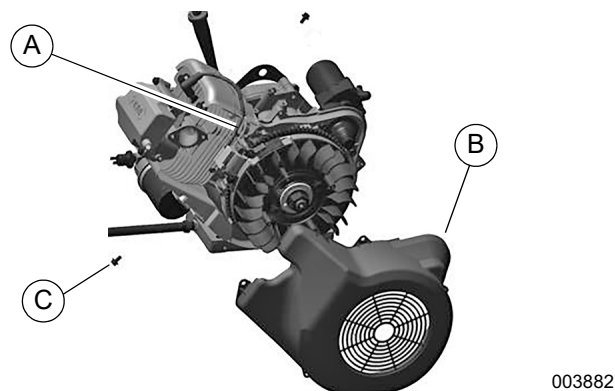


Figure 14-65. Install Blower Housing (HSB)

2. See [Figure 14-66](#). Tighten screws (C) and remaining blower housing screws to 6.6 ft-lbs (9.0 Nm).

Installing Recoil Assembly (Portable)

See [Figure 14-67](#). Align recoil assembly as shown with pull handle at the two o'clock position. Secure with four screws and tighten to 2.2 ft-lbs (3.0 Nm).

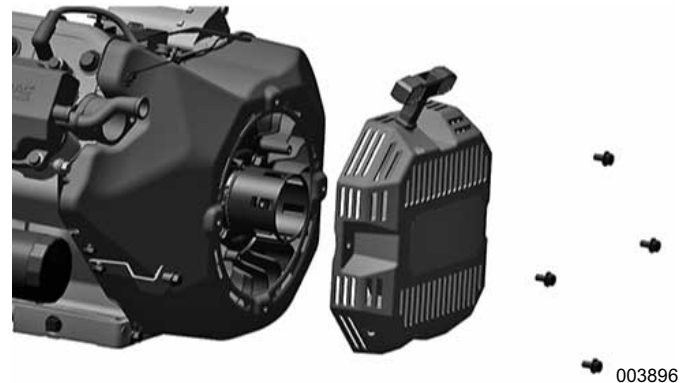


Figure 14-67. Install Recoil Assembly

Installing External Governor and Intake System

See unit diagnostic manual.

Section 15: Specifications

General	
Compression while cranking (three pull minimum—cold engine)	Minimum 55 psi (379 kPa)
Compression ratio	HSB: 9-9.9:1 Portable (gasoline): 7.9-8.8:1
Displacement	28.3 in ³ (459 cc)
Oil capacity (with oil filter change)	35 oz (1.1 qt or 1.03 L)
Oil capacity (without oil filter change)	30 oz (0.95 qt or 0.9 L)
Stroke	426: 2.64 in (67 mm) 460: 2.72 in (69 mm)
Dimensional Information	
Allowable end play	0.005–0.026 in (0.12–0.66 mm)
Balancer bearing bore	1.377–1.378 in (34.967–34.992 mm)
Base circle diameter of cam	1.181 ± 0.0008 in (30 ± 0.02 mm)
Cam lift	Intake: 0.2480 ± 0.0016 in (6.30 ± 0.04 mm) Exhaust: 0.2480 ± 0.0016 in (6.30 ± 0.04 mm)
Camshaft bearing bore diameter	1.083–1.084 in (27.500–27.530 mm)
Compression release lift (measured at tappet)	0.0394 in (1 ± 0.06 mm)
Crankpin diameter	1.416–1.417 in (35.980–36.00 mm)
Crankpin bearing diameter	1.419–1.420 in (36.060–36.070 mm)
Cylinder bore diameter	426cc: 3.543–3.544 in (90.000–90.025 mm) 460cc: 3.622–3.623 in (92.000–92.025 mm)
Exhaust valve stem diameter	0.2766 -0.0016/-0.0022 in (7 -0.040 mm/-0.055 mm)
Flywheel journal diameter	1.3780–1.3784 in (35.0020–35.0130 mm)
Flywheel side journal diameter	1.0802–1.0810 in (27.438–27.458 mm)
Force required to compress spring to 1.25 in (31.8 mm)	16.15 ± 5% lbs (71.8 ± 5% N)
Governor arm bore diameter	0.239 ± 0.003937 in (6.08 ± 0.1 mm)
Governor arm diameter	0.236 ± 0.012 in (6 ± 0.05 mm)
Governor gear shaft diameter	0.236 +0.0004724/+0.0001575 in (6 +0.012/+0.004 mm)
Inner gerotor bore	0.354 +0.0007/-0 in (9 +0.019/-0 mm)
Inner gerotor shaft outside diameter	0.354 -0.0005/-0.0012205 in (9 -0.013/-0.031 mm)
Inner gerotor thickness	0.472 +0/-0.0020 in (12 +0/-0.05 mm)
Intake valve stem diameter	0.276 -0.0008/-0.0014 in (7 -0.02/-0.035 mm)

Large end inside diameter	1.42 in (36.07–36.06 mm)
Main bearing bore diameter (flywheel end)	2.835–2.835 in (72.000–72.0190 mm)
Main bearing diameter (PTO end)	1.379–1.381 in (35.0440–35.0650 mm)
Main camshaft bearing diameter (flywheel end)	1.08 -0.0017/-0.0024 in (27.5 -0.042/-0.062 mm)
Main camshaft bearing diameter (PTO end)	1.299 -0.0020/-0.0028 in (33 -0.05/-0.07 mm)
Oil control ring groove width	0.118 +0.0022/+0.0012 in (3.0 +0.055/+0.030 mm)
Oil control ring width	0.102–0.109 in (2.603–2.756 mm)
Oil pressure relief valve spring: force required to compress spring to 0.7 in (17.8 mm)	2.06 ± 0.21 lbs (9.15 ± 0.92 N)
Outer gerotor shaft outside diameter	1.299 -0.0020/0.0031 in (33 -0.05/-0.08 mm)
Outer gerotor thickness	0.472 +0.0010 in (12 +0.025/0 mm)
Piston major diameter	426cc: 3.543 ±0.0003 in (90 ±0.008 mm) 460cc: 3.622 ±0.0003 in (92 ±0.008 mm)
Piston pin length	1.929 +0/-0.012 in (49 +0/-0.3 mm)
Piston pin outside diameter	0.7872–0.7874 in (19.9950–20.000 mm)
Piston pin bearing diameter	0.7875–0.7879 in (20.005–20.015 mm)
Piston pin bearing bore diameter	0.7879–0.7883 in (20.015–20.025 mm)
PTO journal diameter (crankshaft)	1.3759–1.3775 in (34.950–34.9910 mm)
PTO journal diameter (camshaft)	1.2964–1.2972 in (32.930–32.950 mm)
Pump tip clearance	0.003–0.008 in (0.076–0.20 mm)
Piston ring end gap (top)	426cc: 0.010–0.020 in (0.25–0.50 mm) 460cc: 0.007–0.017 in (0.17–0.34 mm)
Piston ring end gap (center)	0.01–0.02 in (0.25–0.5 mm)
Piston ring end gap (oil)	426cc: 0.0098–0.0197 in (0.25–0.5 mm) 460cc: 0.01–0.03 in (0.25–0.75 mm)
Piston ring side clearance (top and center)	0.015–0.045 in (0.38–1.15 mm)
Piston ring side clearance (oil)	0.0016–0.0035 in (0.04–0.09 mm)
Small end inside diameter	0.0005–0.007 in (0.012–0.18 mm)
Tappet diameter (intake and exhaust)	0.003 in (0.076 mm)
Center ring width	0.1358–0.1457 in (3.45–3.70 mm)
Top and center ring groove width	0.059 +0.0022/+0.0012 in (1.5 +0.055/+0.030 mm)
Top ring width	0.134–0.142 in (3.4–3.6 mm)
Valve clearance (intake and exhaust)	0.002–0.004 in (0.05–0.1 mm)
Valve guide bore diameter (intake and exhaust)	0.2765–0.2763 in (7.005–7.020 mm)
Valve margin	Intake: 0.020 in (0.52 mm) Exhaust: 0.014 in (0.35 mm)
Valve seat angle	44 ± 0.25°

Valve seat width	Intake: 0.039 in (1.0 mm) Exhaust: 0.059 in (1.5 mm)
Valve springs—free length	1.496 in (38 mm)
Valve tappet bore diameter	0.354 +0.0008/-0.0002 in (9 +0.020/-0.006 mm)
Wrist pin bore diameter	0.788 ± 0.0002 in (20.01 ± 0.005 mm)
Torque Values	
Connecting rod bolts	18.6 ft-lbs (24.4 Nm)
Cylinder head bolts	30.2 ft-lbs (41.0 Nm)
Fan plate screws	18.1 ft-lbs (24.5 Nm)
Flywheel nut	90.0 ft-lbs (122.0 Nm)
Gear cover screws (9)	20.0 ft-lbs (27 Nm)
Ignition coil screws	6.6 ft-lbs (9.0 Nm)
Oil drain fitting	10.0 ft-lbs (13.5 Nm)
Oil filter	5.2 ft-lbs (7.1 Nm)
Oil plugs (drains)	10.0 ft-lbs (13.5 Nm)
Oil pressure switch	7.0 ft-lbs (9.5 Nm)
Oil pump pick-up fasteners	6.6 ft-lbs (9.0 Nm)
Oil fill tube fastener	6.6 ft-lbs (9.0 Nm)
Recoil starter screws	2.2 ft-lbs (3.0 Nm)
Rocker arm jam nut	10.0 ft-lbs (13.5 Nm)
Rocker cover screws	6.6 ft-lbs (9.0 Nm)
Rocker studs	9.2 ft-lbs (12.5 Nm)
Shroud fastener	6.6 ft-lbs (9.0 Nm)
Spark plug	18.4 ft-lbs (25 Nm)
Starter motor screws	20.3 ft-lbs (27.5 Nm)
Thermal switch	8.9 in-lbs (1.0 Nm)
Blower housing screws	6.6 ft-lbs (9.0 Nm)

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